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PALEOMAGNETISM AND TECTONICS OF MALAITA, SOLOMON ISLANDS

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Abstract. Malaita, in the Solomon Islands, is the emergent expression of the Pacific Province, a geologically distinct suspect terrane which is commonly held to be the margin of the Ontong Java Plateau, a thickened portion of Pacific plate overthrust onto the rest of the Solomon Islands Arc during a late Miocene reversal of the polarity of the arc. Subduction of the Australian plate replaced Pacific plate subduction during this event, and the Solomon Islands Arc was transferred from the Australian plate to the Pacific plate. A paleomagnetic survey of the eastern Solomon Islands included sampling in southern Malaita. Magnetostratigraphic dating gives an age of 5.8 Ma for the Haruta Calcisiltite-Hada Calcisiltite facies change on Malaita. This facies change resulted from renewed volcanic activity following the initiation of Australian plate subduction. Paleomagnetic poles from two pre-Pliocene units do not lie near the Pacific apparent polar wander path (APWP), as the conventional model would suggest, but rather fall on the Australian APWP, suggesting that the Pacific Province developed as a part of the Australian plate. Correction for 6.5 Ma of Pacific-Australia relative motion produces an improved, very close fit to the Australian APWP, consistent with the transfer of Malaita, along with the rest of the arc, to the Pacific plate at about that time. Magnetic paleolatitudes from a DSDP hole on the Ontong Java Plateau, which disagree with usually accepted Pacific Tertiary motion, also fail to match the Malaitan data; the possibility that Malaita could be a part of an Ontong Java sub-plate, somehow detached from the rest of the Pacific plate, can be discounted. Malaita and the rest of the Pacific Province arose in the fore-arc of the originally north-east facing arc. Collision of the Ontong Java Plateau with the trench slowed subduction of the Pacific plate and began subduction of the Australian plate below the arc. Both subduction zones may have been active for a period, beginning at about 10 Ma and ending at about 3 Ma, until Pacific plate subduction broke down, and the Pacific Province was subsequently uplifted and deformed.

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INTRODUCTION

Malaita lies near the south-eastern end of the Solomon Islands Arc, which lies east of Papua New Guinea and forms part of the complex boundary between the Pacific and Australian plates which characterizes the south-west Pacific region (Figure 1). The geology of Malaita differs markedly from that of the neighboring large islands of the Solomon Islands Arc. Consequently, the interpretation of the origin of Malaita has played a key role in the formulation of models of the tectonic history of the Solomons. Contrasts between Malaita and the rest of the arc have been taken to indicate a tectonically distinct origin for this island, with the suggestion that it represents part of an allochthonous accreted terrane. It was the desire to have an independent means to test this hypothesis that prompted Hackman [1980] to state: "Comparative palaeomagnetic studies of Guadalcanal and Malaita could be very revealing". To this end, 59 paleomagnetic sites were drilled in Malaita, as part of a larger paleomagnetic study of the eastern Solomon Islands which included sampling in Guadalcanal, the Florida Islands, and the Santa Cruz Group.

GEOLOGY OF THE SOLOMON ISLANDS ARC

Coleman [1966] divided the Solomon Islands Arc into three geological zones; the Pacific, Central and Volcanic Provinces. These are shown in Figure 2, following some amendments to their boundaries by Hackman [1973]. Their geology may be summarized as follows [Hackman, 1973, 1980; Coleman and Hackman, 1974; Coulson and Vedder, 1986].

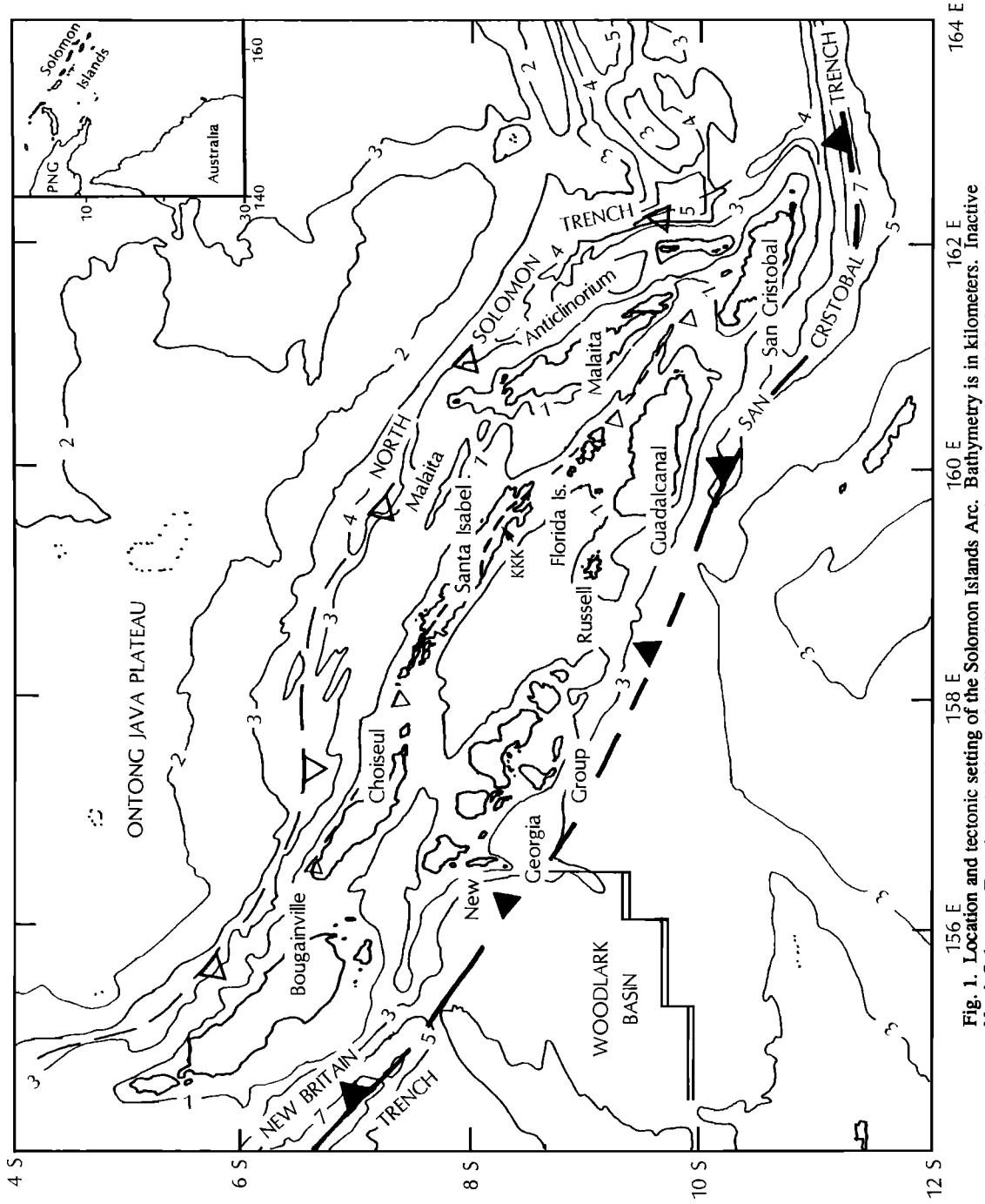


Fig. 1. Location and tectonic setting of the Solomon Islands Arc. Bathymetry is in kilometers. Inactive North Solomons Trench marked by dashed line with large open triangles. Active subduction zone marked by filled triangles, on a solid line where subduction is marked by a trench, and on a dashed line where subduction of the Woodlark Spreading Ridge has suppressed the development of a trench. KKK, Kia-Korigole-Kaipito fault system. Small open triangles mark possible eastern extension of KKK.

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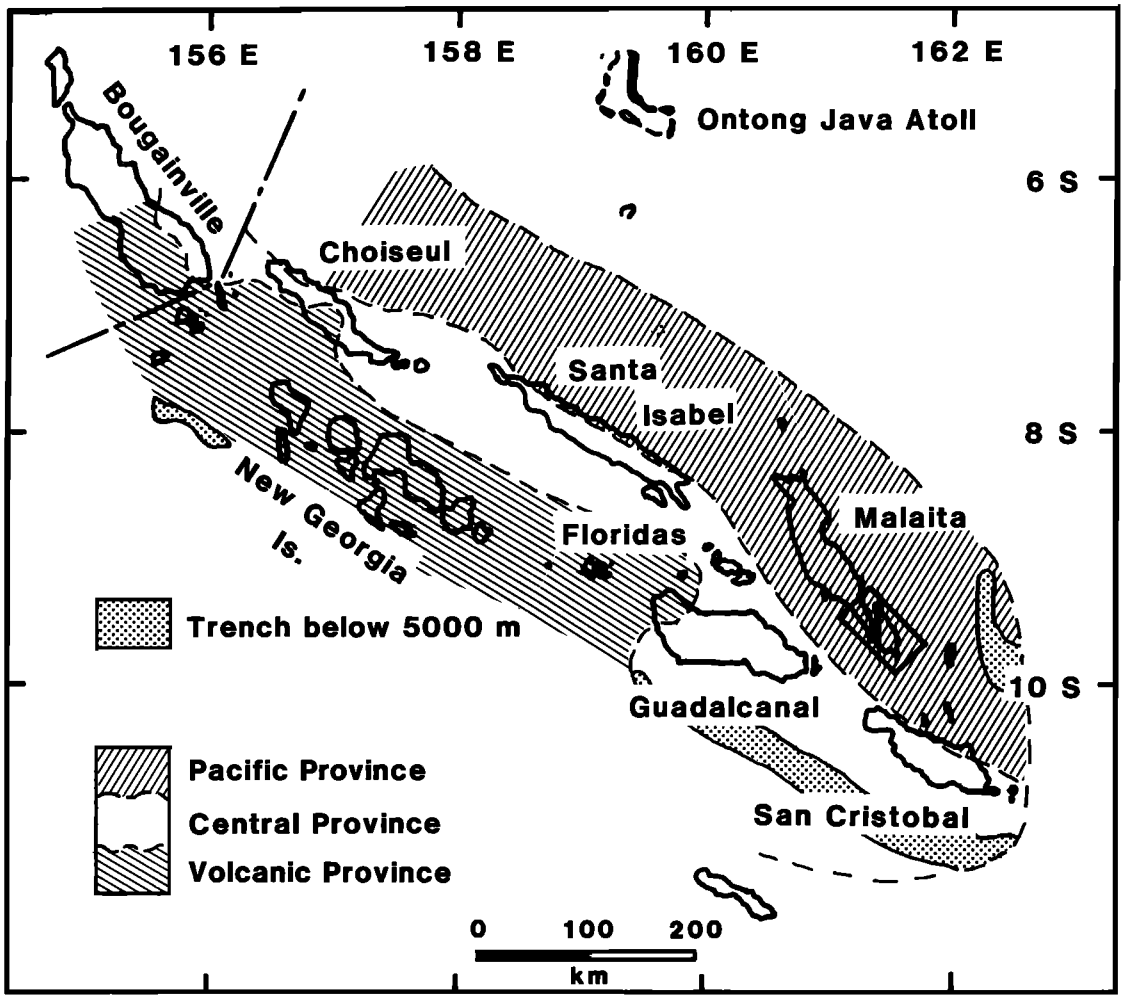


Fig. 2. Geological provinces of the Solomon Islands. The small rectangle encloses the sampling region.

The Pacific Province, which includes the whole of Malaita, the smaller island of Ulawa, and the north-east flank of Santa Isabel (with perhaps some small sections of San Cristobal and Choiseul [Coleman, 1970]), has a core of tholeiitic basalt which displays little metamorphism. This is conformably overlain by predominantly pelagic oozes and marls of Cretaceous to Recent age. Folding occurs along north-west trending axes, with faulting being common and having predominantly the same orientation.

The Central Province contains the greater parts of Guadalcanal, Choiseul and San Cristobal, together with a part of Bougainville and the whole of the Florida Group. The basement of the Central Province includes at least 2000 m of pre-late Eocene (?) mafic lavas and gabbro and diabase intrusions; unlike the Pacific Province these have been subjected to extensive metamorphism of low to intermediate grade (greenschist and amphibolite facies). This metamorphism, occurring during the late Eocene to early Oligocene and accompanied by an unconformity in sedimentation, may mark the initiation of subduction along the north-eastern margin of the arc [Coulson and Vedder, 1986]. Gabbroic and ultramafic material was intruded in the late Cretaceous to early Tertiary. Late Oligocene to Holocene limestones, calcarenites, turbidites and volcaniclastics form an unconformable cover with a maximum thickness of about 5000 m developed in Guadalcanal. Intermediate to acid volcanics and intrusives of late Oligocene to early Miocene and late Miocene to Pleistocene age also occur [Pound, 1986].

The Volcanic Province comprises the presently active arc. It is centered on the New Georgia Group and is characterized by Pliocene to Holocene basaltic and andesitic cones and their surrounding epiclastic apron.

The boundary between the Volcanic and Central Provinces is gradational, but the on-land boundary between the Central and Pacific Provinces has been chosen to coincide with the Kia-Korigole-Kaipito (KKK) fault system in Santa Isabel [Coleman and Hackman, 1974]. The implication that the Pacific Province may be fault bounded along its entire southern margin is consistent with a view of the Central and Pacific Provinces as formerly distinct tectonostratigraphic terranes.

PLATE TECTONIC MODELS FOR THE EVOLUTION OF THE ARC

Seismicity indicates that the Solomon Islands Arc currently lies on the Pacific plate, with subduction of the India-Australia and Solomon Sea plates occurring below the arc

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along a Benioff zone dipping north-east [Luyendyk et al., 1973; Weissel et al., 1982]. Polarity reversal of the Solomons Arc in the late Miocene (Figure 3) has been the consensus of many authors, e.g., Karig and Mammerickx [1972], Halunen and von Herzen [1973], Packham [1973], Falvey [1978] and Kroenke [1984]. Polarity reversal is a process in which subduction below an island arc ceases, the arc is accreted to the formerly consumed plate (here, the Pacific plate), and subduction reverses direction to consume the formerly over-riding plate. An analogous history has been proposed for the New Hebrides (Vanuatu) Arc.

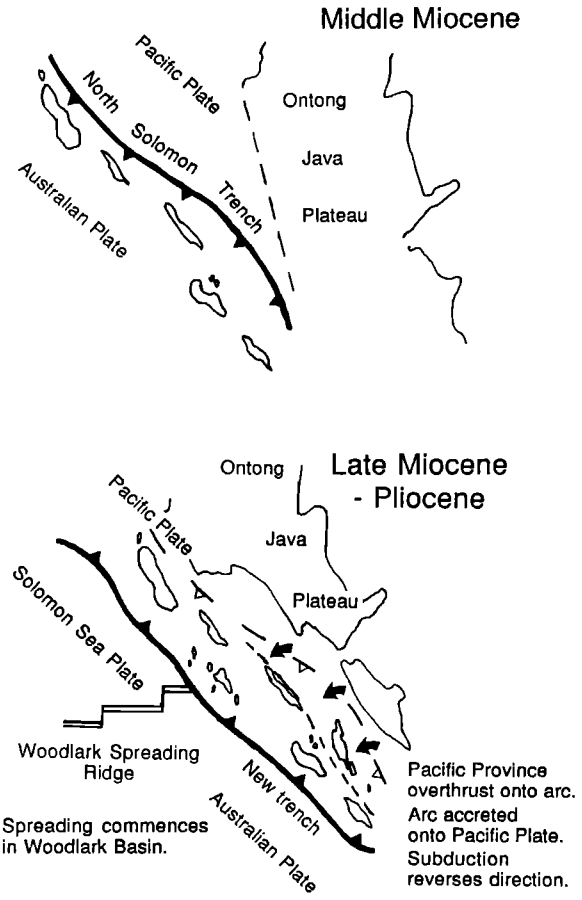


Fig. 3. Commonly accepted model for the tectonic evolution of the Solomon Islands Arc. Thickest, central part of the Ontong Java Plateau is shown surrounded by 2 km isobath (dashed south-western boundary has since been overthrust or consumed).

Seismic evidence [Curtis, 1973; Cooper and Taylor, 1985, 1987] supports the concept of a polarity reversal of the Solomons Arc. Halunen and von Herzen [1973] suggested that the group of isolated deep earthquakes below the Solomons may be explained by the descent of a disjoint piece of lithosphere from the north-east, as a broken fragment of formerly consumed Pacific plate.

Immediately north of the Solomons Arc lies the Ontong Java Plateau (OJP), a large region of thickened Pacific crust whose presence suggests that its collision with the former subduction zone may have been the cause of the polarity reversal in the Solomon Islands Arc [Kroenke, 1972, 1974, 1984; Packham, 1973]. Subduction of the Pacific plate may have been "choked" by the excess buoyancy of the thick crust of the plateau, and subduction of the Australian plate would then have commenced south-west of the Solomons in response to continued Pacific-Australia convergence. Falvey [1978] considered that polarity reversal in the New Hebrides Arc was a similar response to collision of the subduction zone with buoyant, thick oceanic crust, in this case the Melanesian Border Plateau. In his scheme, reversal in the Solomon Islands Arc occurred at about 10 Ma, about 4 Ma earlier than in the New Hebrides Arc.

The Pacific Province has been interpreted as the uplifted edge of the OJP, emplaced at the time of collision [Kroenke, 1972; Halunen and von Herzen, 1973; Coleman and Packham, 1976]. Thrust faults and ultramafic pods mark parts of the boundary between the Pacific and Central provinces, and these were thought to have resulted from the low angle obduction of the OJP over the Central Province [Kroenke, 1972]. Support for this proposition came from correlation of the successions found in Malaita with strata sampled in DSDP boreholes at site 64 [Kroenke, 1972] and sites 288 and 289 [Hughes and Turner, 1976, 1977] (Figure 4).

Polarity reversal in the late Miocene accompanied by the amalgamation of the Pacific and Central Province terranes has become the broadly accepted model. Variations on this theme have been introduced by several authors. Coleman [1966] suggested that the Pacific Province had originated to the east of its current position, and had been translated westwards along transcurrent faults in Choiseul and Santa Isabel from late Miocene times onward. Later [Coleman, 1975], he pointed to anomalous features of the island chain, and questioned whether the Solomons may have originated as a "leaky transform" rather than an island arc. A highly oblique, shearing collision between the arc and the OJP was a corollary of this proposition [Coleman and Packham, 1976]. A revised estimate of the age of the arc polarity reversal was made by Coleman and Kroenke [1981], who proposed that obduction of the OJP and the beginning of the flip in subduction occurred at about 8 Ma, with uplift of Malaita due to rebound of the overthrust after 6 Ma. In order to account for a lack of volcanism in the Solomons from the latter half of the early Miocene until the late Miocene, Kroenke [1984] suggested a 10 Ma delay between the time when collision with the OJP brought subduction of the Pacific plate to a halt (which he placed in the early Miocene) and the time when subduction of the Australian plate commenced (Figure 5). Kroenke envisaged that the Solomons Arc remained on the Pacific plate during this period, while Pacific-Australia convergence was taken up elsewhere in the region. Bruns et al. [1986a] inferred the continuity of the onshore Malaitan sequence into the Indispensable Basin, between Malaita and a postulated eastern extension of the KKK fault system. Bruns et al. [1986b] identified the KKK system as the leading edge of the obducted OJP slab, and noted the absence of subduction related features in a seismic profile across the North Solomons Trench, implying that subduction formerly occurred along a trench which lay south of the Pacific Province. If so, the original fore-arc basin of the former north-east facing arc must now be absent, and may have been tectonically eroded and covered by the obducted OJP or removed along transforms [Bruns et al., 1986a].

The assumption that the Pacific Province is an integral part of the OJP has been central to the arguments proposed in a number of papers [e.g., Nixon and Coleman, 1978; Coleman et al., 1978; Kroenke, 1984]. Less commonly, this

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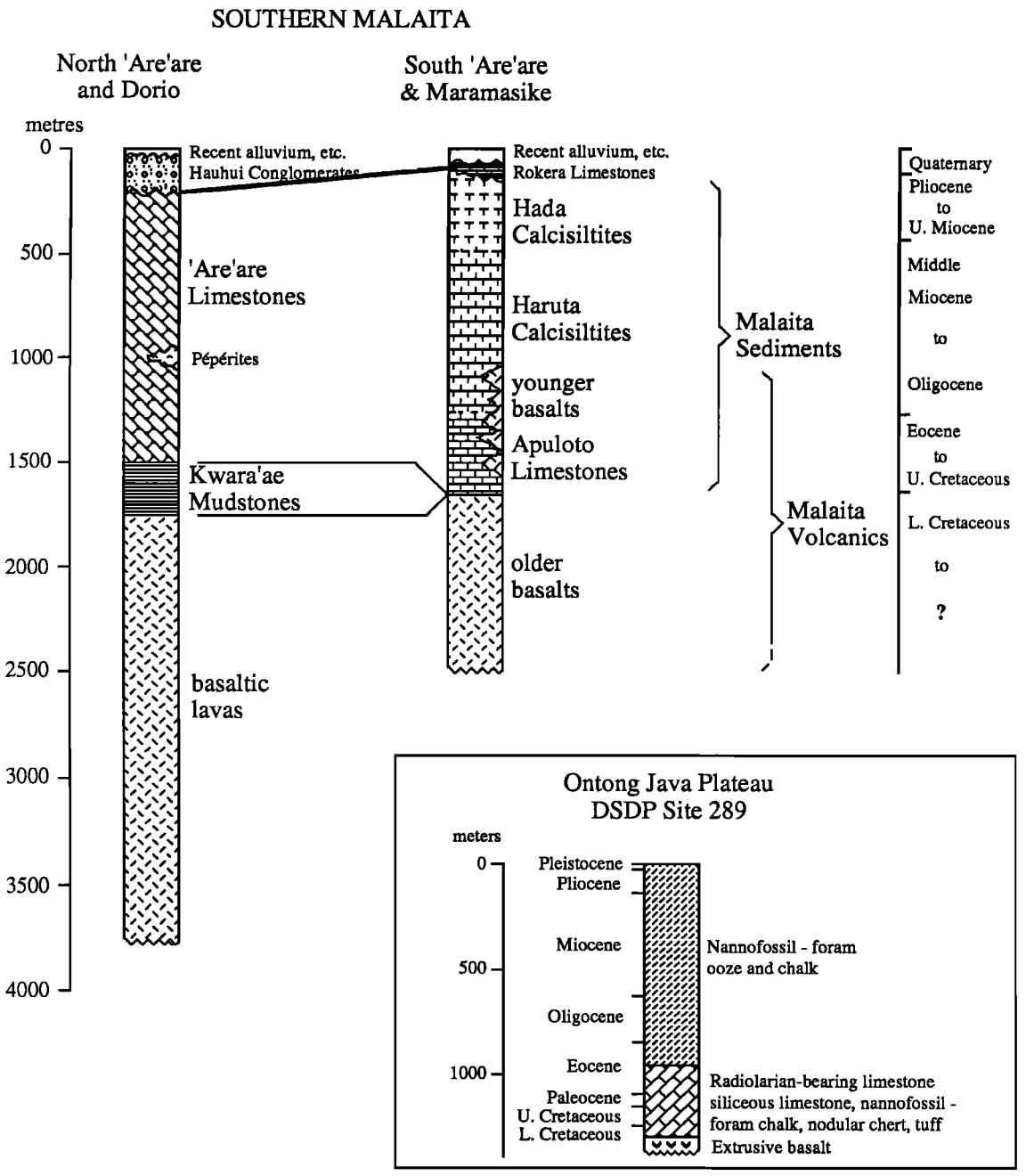


Fig. 4. Stratigraphic successions in two districts of southern Malaita and a simplified column for DSDP site 289, after Hughes and Turner [1976]. Paleomagnetic sampling was carried out in the districts of South 'Are'are and Maramasike (Small Malaita); North 'Are'are and Dorio adjoin to the north-west. The ages indicated on the right, which refer to the South 'Are'are and Maramasike column, are those suggested by Hughes and Turner [1976]. DSDP site 289 is typical of drill sites on the Ontong Java Plateau.

proposition is questioned. Ramsay [1978, 1982] reported observations of structural features from Choiseul and Santa Isabel, which he combined with sedimentological and geochemical evidence from the whole arc to infer that the Pacific Province was not the result of obduction. Ramsay concluded that the basement rocks of ophiolitic type exposed in many of the major islands had always been contiguous, and that the basement of the whole Solomon Islands Arc (both the Pacific and Central Provinces) represents the western edge of the OJP. No lithosphere had ever intervened between the OJP

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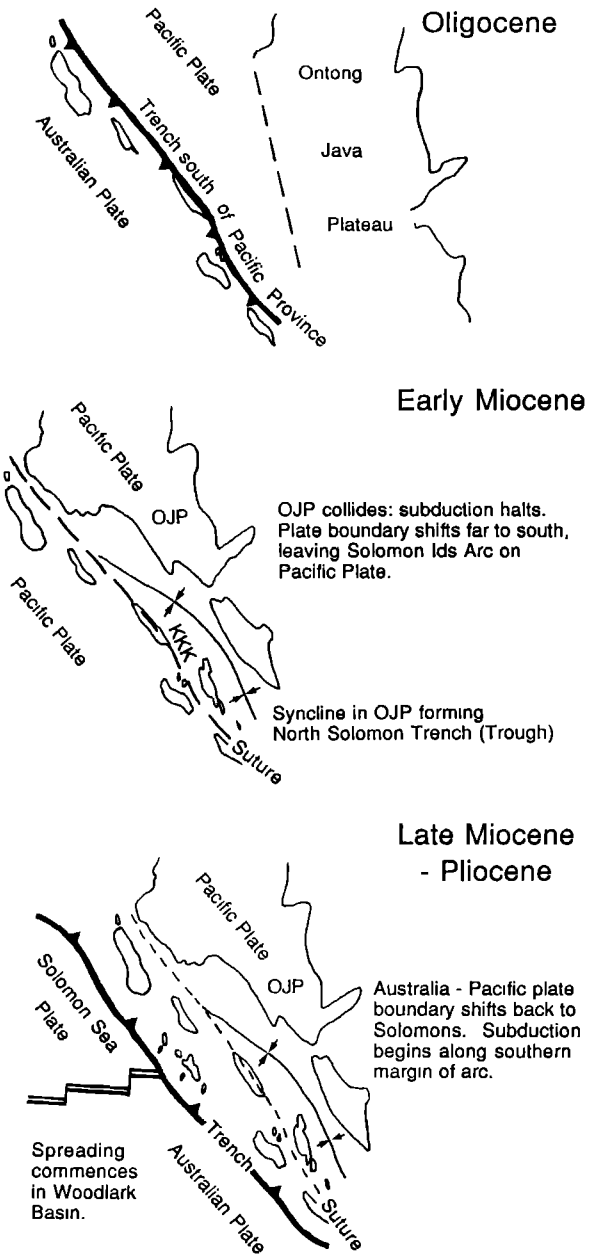


Fig. 5. Alternative model for tectonic evolution of Solomon Islands Arc, after Kroenke [1984] and Bruns et al. [1986b].

and the arc itself, and so no subduction from the north had occurred.

Turner and Hughes [1982] emphasized the presence in both the Pacific and Central provinces of similar sedimentary facies from the Cretaceous to the present, and discounted any differences in style. They concluded that there was no evidence that the pre-Oligocene sediments of the Pacific Province had been deposited far from the Central Province, countering the suggested allochthonous origin for the Pacific Province. Their model implied that the arc had always faced south-west, negating the need for an arc polarity reversal.

THE GEOLOGY OF MALAITA

The stratigraphic succession in southern Malaita, as determined by Hughes and Turner [1976], is shown in Figure 4. The pre-Pleistocene geology of Malaita can be split into two broad divisions; the basaltic Malaita Volcanics, which are present as both pillowed and massive flows and as occasional dykes in the basement and in the lower part of the sedimentary sequence, and the predominantly calcareous, pelagic Malaita Sediments, with a maximum thickness of about 1500 m.

In the southern part of Malaita basaltic flows occur within the lower part of the Malaita Sediments; these are termed the "younger basalts" to distinguish them from the "older basalts" of the basement. Hughes and Turner [1976] identify the older basalts on both mineralogical and chemical grounds as tholeiites, and suggest that the "younger basalts" should probably be termed alkali olivine basalts.

In the south of the island the Malaita Sediments are comprised solely of the 'Are'are Limestones. Distinct though gradational divisions can be seen within the 'Are'are Limestones, bounding the three units termed by Hughes and Turner [1976] the Apuloto Limestones, the Haruta Calcisiltites, and the Hada Calcisiltites.

The Apuloto Limestones, well recrystallized (sometimes porcellanous) white limestones, form the base of the 'Are'are Limestones. Much of the unit is massive; where bedding does occur it is marked either by chert bands or by sharp breaks which are difficult to distinguish from jointing. Paleomagnetic sampling was consequently limited by lack of structural control.

The Apuloto Limestones are overlain by the Haruta Calcisiltites. Creamy-white to pale grey calcisiltites make up the bulk of the unit; intercalated bands of brown mudstone, typically 5 to 20 cm thick and separated by about 15 to 30 cm of calcisiltite, mark the bedding. Bioturbation commonly destroys any sedimentary features present in the mudstones; where this is less intense the bands show features typical of distal turbidites.

The Haruta Calcisiltites grade into the overlying Hada Calcisiltites. These latter consist predominantly of blue-grey calcisiltites and calcilutites, the colour rapidly altering to cream upon drying. The calcisiltites are soft and highly porous. They are often well bedded. More active environmental conditions are indicated by frequent calcarenites and pebbly mudstones. Brown mudstone bands up to a few centimeters in width occur in places; the thin, turbiditic mudstone bands of the Haruta section are far less common in the Hada Calcisiltites. Although petrographically similar to the Haruta Calcisiltites, the Hada Calcisiltites contain considerably more terrigenous detritus.

Earlier work on planktonic foraminifera from the 'Are'are Limestones was assembled by Hughes and Turner [1976] to provide the following age ranges: the Apuloto Limestones from late Cretaceous to late Eocene; the Haruta Calcisiltites from Oligocene to middle Miocene; and the Hada Calcisiltites from late Miocene to late Pliocene/Pleistocene. Samples collected as part of the paleomagnetic study were also dated by D. Belford (personal communication, 1984), again based on foraminifera. Foraminiferal zones to which sites were assigned are given in Table 1.

The change in facies between the Haruta and Hada Calcisiltites (and their equivalents in northern Malaita, the Suaba Chalks and Tomba Silts of Rickwood [1957]), particularly the increase in the terrigenous component and what has been interpreted as evidence of shallowing [Hughes and Turner, 1976], have been taken to reflect events connected with the collision of the OJP with the arc and the ensuing arc

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TABLE 1. Foraminiferal Ages of 'Are'are Limestones

Site(s)	Zone
Hada Calcisiltites	
SI 57 - 74 & SI 81	N21
SI 75	N19-N21
SI 76 - 80 & SI 82	N19
Haruta Calcisiltites	
SI 83	Older than N19; 15 m stratigraphically below SI 82, so N17 or N18.
SI 84	Not later than N3
SI 85	Not older than N6
SI 93-97 & SI 99-100	Not later than N3
SI 112-114	N3
Apuloto Limestones	
SI 91 P12	

Ages are based on suggestions by D. Belford (personal communication, 1984). Zonal divisions follow Berggren *et al.* [1985a], Berggren *et al.* [1985b] and Blow [1969].

polarity reversal [Nixon and Coleman, 1978; Coulson and Vedder, 1986]. On this basis the arc polarity reversal has usually been placed in the early part of the late Miocene, at around 10 Ma. Foraminiferal paleobathymetric data, however, indicate that significant uplift of the Pacific Province did not commence until about 2.5-3.0 Ma [Resig et al., 1986]. An increased influx of volcanogenic sediments, uplift of the Pacific Province and deformation of the Malaita Anticlinorium were evidently distinct phases in a protracted response to arc polarity reversal, the first of which was related to a renewal of volcanism in the Central Province following the beginning of subduction of the Australian plate.

Radiometric dating of the Malaita Volcanics has not been carried out. Their relationships with the Apuloto Limestones suggest a Cretaceous age for the older basalts, and an approximately Eocene age for the "younger basalts". Hughes and Turner [1976] reported locations where the Haruta Calcisiltites overlie lavas of the "younger basalts", suggesting that their extrusion continued until the late Eocene at least. Alnoitic intrusions dated to 34 Ma occur in north-central Malaita [Nixon and Coleman, 1978], but have not been observed in southern Malaita. Coleman and Kroenke [1981] suggested that intrusion of part of the alkalic "younger basalts" suite occurred at about 10 Ma; Kroenke [1984] amended this to 20 Ma. At least some of the "younger basalts" of southern Malaita are clearly overlain by Apuloto Limestones, and Hughes and Turner [1977] indicate that their extrusion did not continue after the Apuloto-Haruta transition; ages much younger than the late Eocene for the any part of the "younger basalts" are unlikely.

Faults in Malaita are frequent, being mainly either normal or high-angle reverse; thrust faults appear to be rare. Folding and faulting appear to be a response to the same NE-SW directed stress. Fold axes are usually near horizontal, and rarely plunge at more than 20°. Declination errors which may arise from the conventional tilt correction [MacDonald, 1980; Tarling, 1983] would be limited to at most about 7°, given typical south Malaitan dips of about 50°.

FIELD SAMPLING

It was hoped to sample throughout the sequence, to allow determination of paleomagnetic poles for each of the lithological divisions and the construction of a low-resolution magnetostratigraphy. Sampling in the lower units was restricted by limitations of outcrop and access; nevertheless, only the older basalts remained essentially unsampled. Five sites were drilled in flows and dykes clearly part of the "younger basalts". The stratigraphic position of a sixth basaltic site (SI 106) was unclear; although indicated on the geological sheet as an outcrop of older basalt, field relationships suggest that it is part of the same layer of "younger basalts" sampled by SI 107 - SI 109, repeated 1.5 km away by an unmapped fault or fold. No other exposures of the older basalts occur in areas accessible to this study. Four sites were drilled in the Apuloto Limestones, including one transitional to the Haruta Calcisiltites. The Haruta Calcisiltites were sampled at 22 sites, including one just at the top of the transition from the Apuloto Limestones. Samples from the Hada Calcisiltites were taken at 27 sites, including three (possibly four) in the 40-50 m transition zone from the Haruta Calcisiltites. The locations of sampled sites are shown in Figure 6.

Each site was sampled by five cores. Cores were oriented by magnetic compass; weather conditions and rainforest cover usually precluded the use of a sun-compass. However, magnetic susceptibilities of all of the rocks sampled were sufficiently low to avoid producing any significant error in the magnetic compass.

ANALYSIS

Laboratory Technique

Stepwise demagnetization was carried out on one 2.5 cm long specimen from each core, by the alternating field (AF) method for the sedimentary specimens and the thermal method for the igneous specimens [Collinson, 1983]. Demagnetization paths for the AF-cleaned sediments were frequently highly noisy, and duplicate specimens were demagnetized by thermal means. Magnetization was measured in most cases on a two-axis SCT cryogenic magnetometer; where intensities exceeded the range of this device, a Digico spinner magnetometer was used. Magnetometers and demagnetizing apparatus were magnetically shielded by Helmholtz coils. Specimens were stored in nearly field-free space throughout their demagnetization and measurement, although some exposure to laboratory fields and to the earth's field did inevitably occur.

Interpretation of demagnetization behavior was aided by the preparation of orthogonal plots [Zijderveld, 1967]. Linear components of magnetization were fitted using a computer program based on the analysis of Kirschvink [1980]; highly noisy demagnetization behavior, especially at higher AF or thermal demagnetization levels, prevented the efficient operation of automatic routines to pick linear segments of the demagnetization path, and so subjective picking of magnetization vectors to be fitted was often required.

Auxilliary studies of some of the magnetic and physical properties of the suite of rocks from Malaita helped to elucidate their magnetic history. Detailed results are reported in Musgrave [1986]. Laboratory-induced isothermal remanent magnetization (IRM) studies [McElhinny, 1973; Achache et al., 1982] showed magnetite to dominate the remanence. Hematite, though present, contributes little to the

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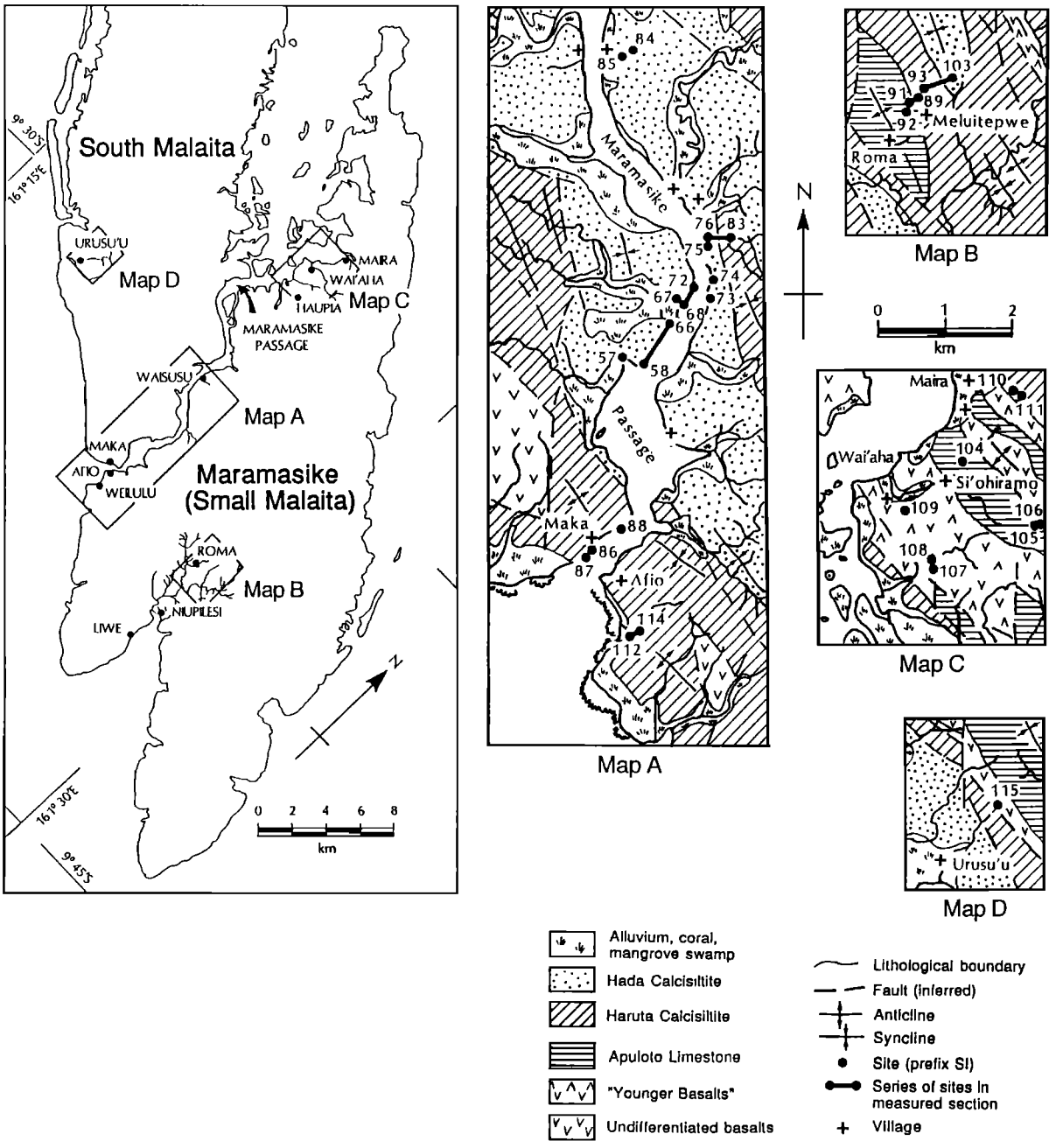


Fig. 6. Location of paleomagnetic sites in southern Malaita. Geology is from 1:50 000 sheets produced by the Solomon Islands Geological Survey. Sites SI 84 and SI 85, although indicated as Hada Calcisiltites on the 1:50 000 sheets, were in fact sampled in a belt of Haruta Calcisiltites of unknown extent.

remanence; most of the hematite evidently has a low coercivity, and much may be superparamagnetic. Rotational remanent magnetization (RRM) [Wilson and Lomax, 1972] was a frequent source of noise in the AF-demagnetized sediments, often obscuring demagnetization behavior above 25 mT. Two-axis tumbling, used in this study, has been shown to be susceptible to the acquisition of RRM [Brock and Iles, 1974]. Studies of the acquisition of viscous remanent magnetization (VRM) showed the sedimentary rocks to be highly susceptible to short-term VRMs, with a relaxation time of a few hours to a few months, and implied also a significant overprinting of the primary magnetization by long-term VRM. X-ray and optical studies [Bouma, 1969; Musgrave, 1986;] confirmed the extensive influence of deformative bioturbation on the sediments. Pelloidal glauconite, a prominent constituent of the Hada and Haruta

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Calcisiltites, may indicate deposition of these units as fecal pellets produced by filter feeders [Pryor, 1975]. Bioturbation and fecal pellet deposition both imply that depositional remanence will have been replaced by post-depositional fields [Verosub, 1977], which will not be affected by any depositional "inclination error".

Demagnetization Behavior

'Are'are limestones. The presence of a large long-term VRM component is clearly visible in the demagnetization paths of many of the carbonate specimens (Figure 7). Similar susceptibility to long-term VRM in carbonates has frequently been reported, and was originally described by Creer [1959]. Stability spectra of the primary magnetization and the VRM under AF demagnetization frequently overlapped to the extent that it was impossible to isolate either component purely. Discrimination of the two components under thermal cleaning was far better; a similar response of a VRM in carbonates has been noted by other workers [e.g., Acache et al., 1982; Scotese et al., 1982]. As the long-term VRM is normally polarized, recognition of reversed primary magnetizations was relatively straightforward, although in some cases the VRM persisted to such high demagnetization levels that isolation of the primary direction was impossible (Figure 8). Normally polarized primary directions contrasted much less strongly with the VRM, particularly in the younger specimens (Hada Calcisiltites) where little difference between the paleo-direction and the present dipole can be expected. In many sedimentary specimens complete, or at least substantial, isolation of the primary magnetization at higher demagnetization levels is suggested by a clear inflection in the demagnetization path at around 20-25 mT or 200-300°C, followed by linear convergence on the origin of the Zijderveld plot above this level (Figure 9). Specimens in which the primary magnetization was not clearly isolated were omitted from calculations of site means. Mis-identification of VRMs as normal primary magnetizations, or imperfect isolation of the primary component, remains a risk in some specimens, however.

A few sites (SI 58, SI 75, SI 81 and SI 102 from the Hada Calcisiltites, and SI 93 from the Haruta Calcisiltites) show an apparent partial or complete change in polarity within the site, although some of these apparent polarity changes may result from an erroneous recognition of the primary magnetization.

Random directions, usually removed by 20 mT or 200°C, were a prominent feature of the Hada Calcisiltites, but were "softer" and proportionally smaller in the other rock units. These presumably arose after sampling, and were the result of short-term VRMs, perhaps compounded in the Hada Calcisiltites by chemical remanent magnetization accompanying the colour change which followed drying of the specimens. Reddening of the sedimentary specimens frequently occurred above about 300°C. While such a colour change is often accompanied by the generation of new ferromagnetic phases, few specimens acquired any significant magnetization "noise".

Malaita volcanics. With the exception of sites SI 106 and SI 115, the demagnetization behavior of the basalts was readily interpreted. The path of the demagnetization vector converges linearly on the origin of the Zijderveld plot above

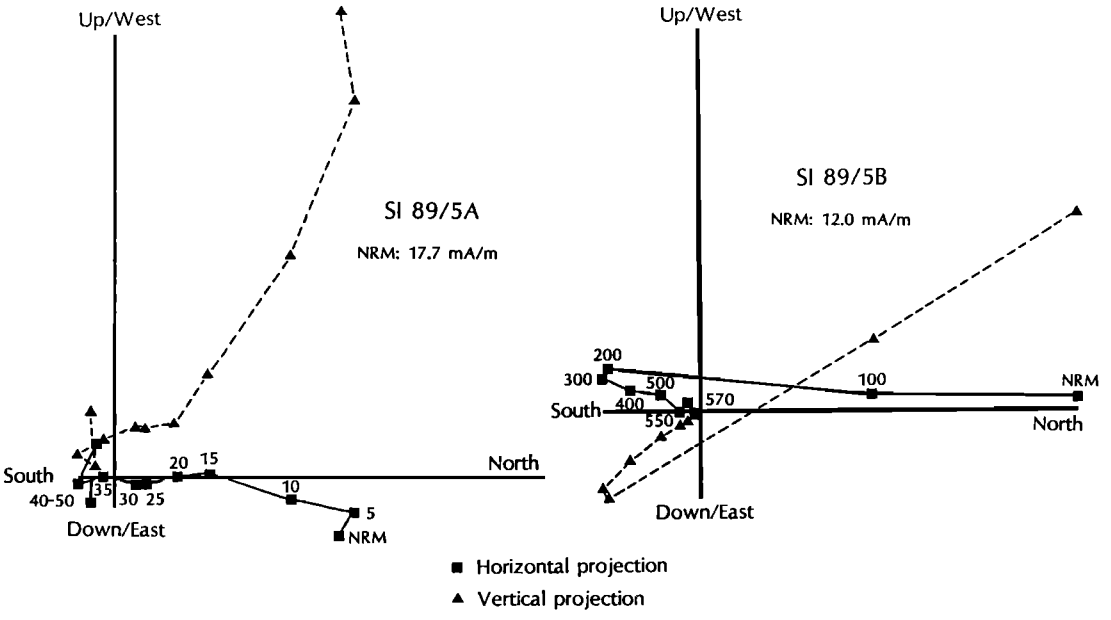


Fig. 7. Zijderveld plot of demagnetization behavior of typical carbonate specimens, here from core 5 of site SI 89 in the Haruta Calcisiltites. As is the case in all subsequent figures, the magnetization directions have not been corrected for bedding. Specimen 5A has been demagnetized by AF (field strengths in milliTesla are indicated for each step), 5B by heating (temperatures in degrees Celsius are indicated). A large, normally polarized component is present as an overprint on a reversed primary magnetization. Overlap of the stability spectra of the two components is much greater under AF demagnetization than under thermal cleaning. RRM is generated by AF fields of 25 mT or more, partly obscuring the primary magnetization.

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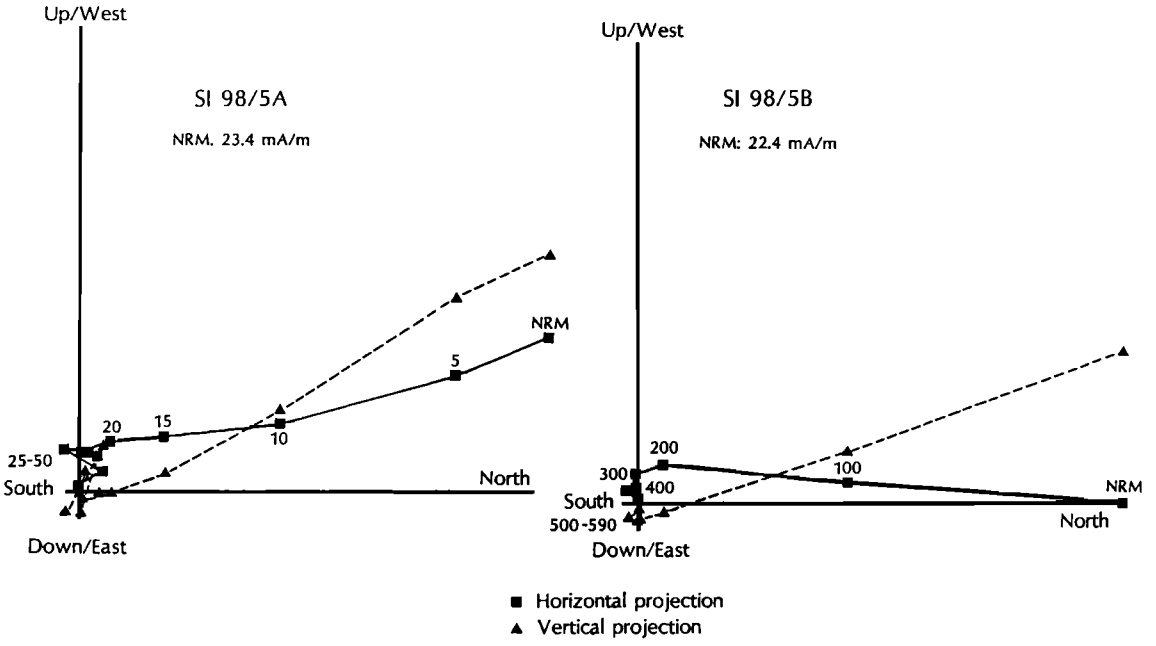


Fig. 8. Zijderveld plot of two specimens from core 5 of site SI 98, drilled in the Haruta Calcisiltites. Specimen 5A has been demagnetized by AF (field strengths shown in milliTesla), 5B by heating (temperatures shown in degrees Celsius). A proportionally large VRM overprint almost completely replaces the reversed primary magnetization. Under AF cleaning, only the failure of the demagnetization path to converge on the origin indicates the presence of the reversed component. Thermal demagnetization reveals the presence of the primary component, but this is never completely isolated from the VRM, and the direction of the primary magnetization cannot be accurately established.

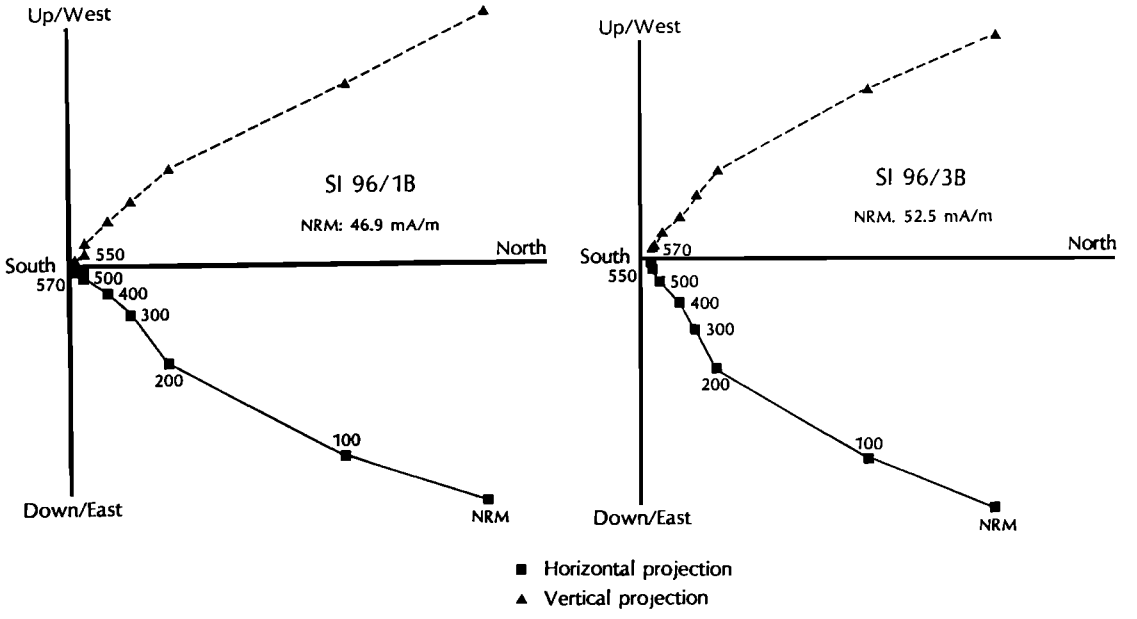


Fig. 9. Zijderveld plot of specimens 1B and 3B, from site SI 96 drilled in the Haruta Calcisiltites. Demagnetization temperatures, in degrees Celsius, are indicated. Thermal demagnetization here reveals the presence of a normal component isolated from the VRM above 200°C. The sharp termination of the VRM stability spectrum is indicated by the clear inflection in the demagnetization path at 200°C, followed by convergence on the origin. Below 200°C, curvature of the demagnetization path indicates overlap of the stability spectra of the primary magnetization and VRM.

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about 500°C, all overprints being removed (Figure 10). Small overprints in many specimens are the result of long-term VRM aligned in the direction of the modern dipole field. Larger, random overprints, removed by 100°C, were acquired after sampling. Polyphase intrusion of the dyke sampled at SI 92 resulted in the acquisition of a stable overprint close in direction to the primary magnetization.

Specimens of site SI 106 have an unexpectedly weak magnetization (NRM approximately 10^{-4} to 10^{-3} Am^{-1}); their demagnetization paths are complex and inconsistent. This site was sampled in a relatively thick (greater than 30 cm) layer of sideromelane, partially palagonized, and resembling an occurrence noted by Hughes and Turner [1976]. The unusual petrology may account for the complex magnetization. Site SI 115, although certainly basaltic, nevertheless behaves inconsistently from specimen to specimen; demagnetization paths are simple, but differ in direction from one to another. Lightning-induced IRM seems the most likely origin for this magnetization.

INTERPRETATION

Where cores were represented by duplicate AF and thermal demagnetization, the representative magnetization was taken from the treatment whose most stable ("characteristic") component was best isolated. Site means of characteristic components, assumed in the first analysis to be "primary", are listed in Table 2, both before and after structural (tilt) correction. Rock unit means are also presented; magnetization directions of reversely polarized sites have been inverted in the calculation.

Several sites which have preserved stable, apparently primary magnetizations were nevertheless omitted from the calculation of rock unit mean directions and poles. Sites SI 58 and SI 61 were omitted due to their lack of reliable local structural information, while SI 75 and SI 102 showed too much internal variation for a mean to be defined. SI 100 was intermediately polarized, and so was also excluded.

Field evidence suggests that site SI 101 in the Haruta Calcisiltites is overturned, and the structural correction was made accordingly. Structural correction of the basalt sites presented a problem. Where it could be determined locally (SI 107, SI 108, SI 109, SI 115) the dip of the basalts was measured from surfaces representing the top of a layer of lava pillows; such a plane may not be an exact original horizontal. The tilt of one sampled dyke (SI 92) was determined by bedding in the surrounding limestones; the other sampled site (SI 106) failed to record a stable magnetization.

Stability Tests

There are a number of tests which may be applied to assess whether characteristic magnetizations are primary, and the extent to which they may have been adulterated by overprints which have been incompletely removed. Least subjective of these is the fold test [Graham, 1949; McElhinny, 1964]. A second test is based on the degree to which normal and reversed directions are antipodal.

The fold test. The fold test of McElhinny [1964] is preferred here to that of McFadden and Jones [1981]. The former test is more strict and is the more appropriate where sites are not restricted to two planar fold limbs, but instead show a continuous distribution of dips, as is the case in

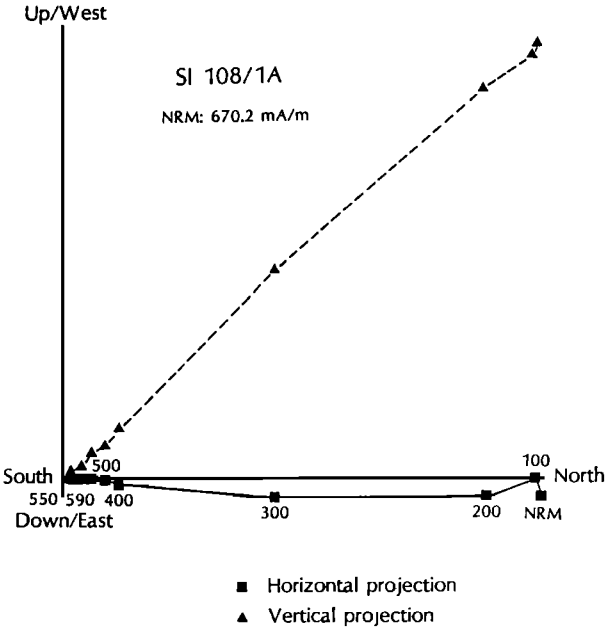


Fig. 10. Zijderveld plot of a typical basalt from the younger basalts of the Malaita Volcanics (specimen 1A from site SI 108). Demagnetization temperatures, in degrees Celsius, are indicated. Thermal demagnetization removes a long-term VRM overprint by 500°C: this overprint, whose stability spectrum overlaps extensively with that of the primary magnetization at lower temperatures, is proportionally small.

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TABLE 2a. Characteristic ("Primary" ?) Magnetizations

Site	Polarity	Uncorrected Dec (°E)	Uncorrected Inc (°S)	Corrected Dec (°E)	Corrected Inc (°S)	N	α95 (°)
			Hada Calcisiltites				
SI 57#	-	-	-	-	-	-	-
SI 58	N/R?	028.7	-38.9	poor structural control		4	14.8
SI 59#	R?	-	-	-	-	-	-
SI 60#	R?	-	-	-	-	-	-
SI 61	R	241.4	32.1	poor structural control		4	23.8
SI 62#	R?	-	-	-	-	-	-
SI 63	R	170.0	13.5	176.8	11.7	4	29.0
SI 64	N	343.6	-28.1	012.0	-8.2	5	7.4
SI 65	N	338.8	-23.3	004.6	-14.2	5	13.4
SI 66	N	004.4	-18.8	002.0	-21.5	5	17.1
SI 67	N	357.9	-28.5	007.8	-15.7	5	5.8
SI 68	N	005.9	-22.3	011.3	-17.1	4	20.6
SI 69	N	250.2	-28.6	354.6	-21.3	5	6.7
SI 70	N	007.5	-26.7	013.0	-19.1	5	17.6
SI 71	N	016.6	-20.5	021.0	-4.7	5	19.7
SI 72	N	012.5	-22.5	018.9	-9.0	5	24.5
SI 73	R	174.3	15.4	183.2	9.0	4	21.2
SI 74	R	172.1	17.3	182.5	12.3	4	19.2
SI 75	I/R/N	wide variation in magnetization direction within site					
SI 76	R	172.1	33.6	191.1	9.7	5	19.7
SI 77	R	159.3	14.6	172.0	14.0	5	11.9
SI 78	R	155.2	28.3	181.7	29.6	5	13.2
SI 79	R	170.0	20.7	186.1	13.1	5	14.3
SI 80	N	355.4	-27.2	015.2	-12.1	5	21.8
SI 81	N/R	138.0	36.9	171.6	27.4	4	30.4
SI 82	N	336.1	-32.3	17.8	-21.9	4	9.6
SI 102	N/I/R	wide variation in magnetization direction within site					
			Haruta Calcisiltites				
SI 83	N	321.4	-58.0	010.8	-40.7	5	29.0
SI 84	N	000.9	-76.6	019.5	-4.7	5	4.1
SI 85	N	015.5	-4.0	018.8	11.4	5	6.5
SI 86	N	356.3	-32.5	003.1	-13.0	5	11.2
SI 87	R	150.9	44.1	162.9	35.0	5	12.9
SI 88	N	015.2	-18.6	356.1	-36.3	5	7.1
SI 89	R	205.1	19.8	197.9	17.8	4	29.5
SI 90	R	148.5	32.6	152.8	5.1	2	*
SI 93	N/R	011.7	-18.7	001.4	-30.1	4	10.5
SI 94	N	020.3	-27.9	008.6	-22.4	5	7.4
SI 95	R	180.4	46.2	154.5	46.8	4	5.6
SI 96	N	044.5	-41.7	015.6	-52.7	5	10.5
SI 97	N	016.2	-39.3	333.8	-57.5	4	10.8
SI 98	R	219.7	28.4	211.5	31.9	4	28.5
SI 99	R	221.8	25.5	215.9	43.7	4	24.5
SI 100	I	300.8	-14.2	299.2	7.3	5	11.6
SI 101	N	062.0	22.4	032.0	-45.3	5	10.8
SI 103	N	039.6	7.6	031.5	-24.2	5	12.7
SI 110	N	012.8	-36.2	012.1	-46.2	4	17.3
SI 112#	R?	-	-	-	-	-	-
SI 113	R	169.6	33.0	179.1	23.4	4	15.8
SI 114	N	003.5	-32.1	004.3	-15.2	5	6.4
			Apuloto Limestones				
SI 91	N	053.8	-11.2	042.9	-37.9	5	6.9
SI 104	N	356.5	-35.6	006.0	-36.5	4	4.3
SI 105	N	308.8	-54.3	001.7	-51.9	4	7.4
SI 111	N	357.9	-25.7	017.1	-26.1	4	17.7

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TABLE 2a. (continued)

Site	Polarity	Uncorrected Dec (°E)	Uncorrected Inc (°S)	Corrected Dec (°E)	Corrected Inc (°S)	N	α95 (°)
			Younger basalts				
SI 92	N	025.4	4.4	020.1	-25.8	5	16.1
SI 106#	?	-	-	-	-	-	-
SI 107	N	021.6	-51.1	018.2	-14.4	5	22.1
SI 108	N	015.8	-32.7	354.2	-38.1	5	28.6
SI 109	N	359.9	-39.4	003.5	-48.7	5	5.6
SI 115#	?	-	-	-	-	-	-

N, normal polarity; R, reversed polarity; I, intermediate polarity. A sequence of the type N/R indicates a change of polarity within the site; the uppermost polarity is shown on the left, the lowermost on the right.

* α95 not defined for N < 3.

indicates that a characteristic component (one represented by a linear segment of the demagnetization vector converging on the origin of the Zijderveld plot) could not be isolated from two or more specimens (i.e., the site is "unstable").

TABLE 2b. Means of Structurally Corrected Site Directions

	Dec (°E)	Inc (°)	N	α95 (°)
	Hada Calcisiltites			
unit mean direction	006.6	-15.5	19	4.5
mean of normally polarized sites only	010.9	-15.1	11	5.5
mean of reversely polarized sites only	180.8	15.9	8	7.1
	Haruta Calcisiltites			
unit mean direction (SI 100 omitted)	007.6	-30.5	20	10.1
mean of normally polarized sites only	011.6	-29.8	12	13.5
mean of reversely polarized sites only	181.7	31.4	7	20.9
	Apuloto Limestones			
unit mean direction	017.6	-39.1	4	20.4
	Younger basalts			
unit mean direction	010.1	-32.2	4	21.0
	Eocene			
combined Apuloto Limestones and younger basalts	013.7	-35.7	8	12.0

Malaita. Site mean directions before and after structural correction are compared in Figure 11 and rock unit means in Table 3. The fold test is significant at the following probability levels:

Rock Unit	N	Probability Level
Hada Calcisiltites	19	95.8%
Haruta Calcisiltites	20	91.6%
Apuloto Limestones	4	95.7%
"younger basalts"	4	78.8%
combined Eocene (Apuloto Limestones and "younger basalts")	8	97.9%

With the exception of the "younger basalts", the fold test shows the characteristic magnetization of each unit to be comprised completely or largely of a component acquired prior to deformation (and so probably primary), with a reliability exceeding 90%. The "younger basalts" also show a marked improvement in grouping following structural correction, suggesting a primary origin even if they formally fail the fold test. When the Eocene data is combined the certainty increases to greater than 95%. Considering the high noise level in the demagnetization data, and the rather acute angle between the rock unit mean directions and the predominant fold axis direction (approximately NNW) which restricts the fold test's power of discrimination, this result provides clear evidence of a predominantly primary origin for all the rock unit mean directions.

Consistency of reversals. Divergence between the means of the normally polarized and the inverted reversely polarized sites from a rock unit may indicate the presence of a contaminating overprint. Watson [1956] developed a criterion to test the significance of such a divergence. Slight modifications introduced to this test by McFadden and Lowes [1981] produce little difference in the region of interest, and as Watson's statistic is slightly the larger, its use is the more cautious. A precondition of the test is the requirement that the precision parameters (k) of the normal and reversed sets are not significantly different [McElhinny, 1973].

None of the sites from the Apuloto Limestones or the "younger basalts" were reversely polarized, an imbalance which itself suggests that primary magnetizations may not have been recorded (see the discussion below). No significant difference exists in the value of k for the Haruta Calcisiltites, so Watson's test may be applied. Watson's statistic has a value of 0.42, which is not significant (given $N_{Normal} = 12$, $N_{Reversed} = 7$), even at only a 50% probability level; the assumption that the characteristic magnetizations represent the two polarities of the same dipole field is supported. Precision parameters for the normal and reversed Hada Calcisiltites are also not significantly different. However, this formation fails the reversals test, as Watson's statistic (for $N_{Normal} = 11$, $N_{Reversed} = 8$) equals 4.76; the normal and reversed populations are different at a confidence level of 98.5%. An overprint is clearly present, deflecting the primary directions of magnetization in an asymmetric manner with respect to polarity.

The fact that the characteristic magnetization of the Hada Calcisiltites "passes" the fold test but "fails" the reversals test could suggest that the overprint was acquired prior to (or

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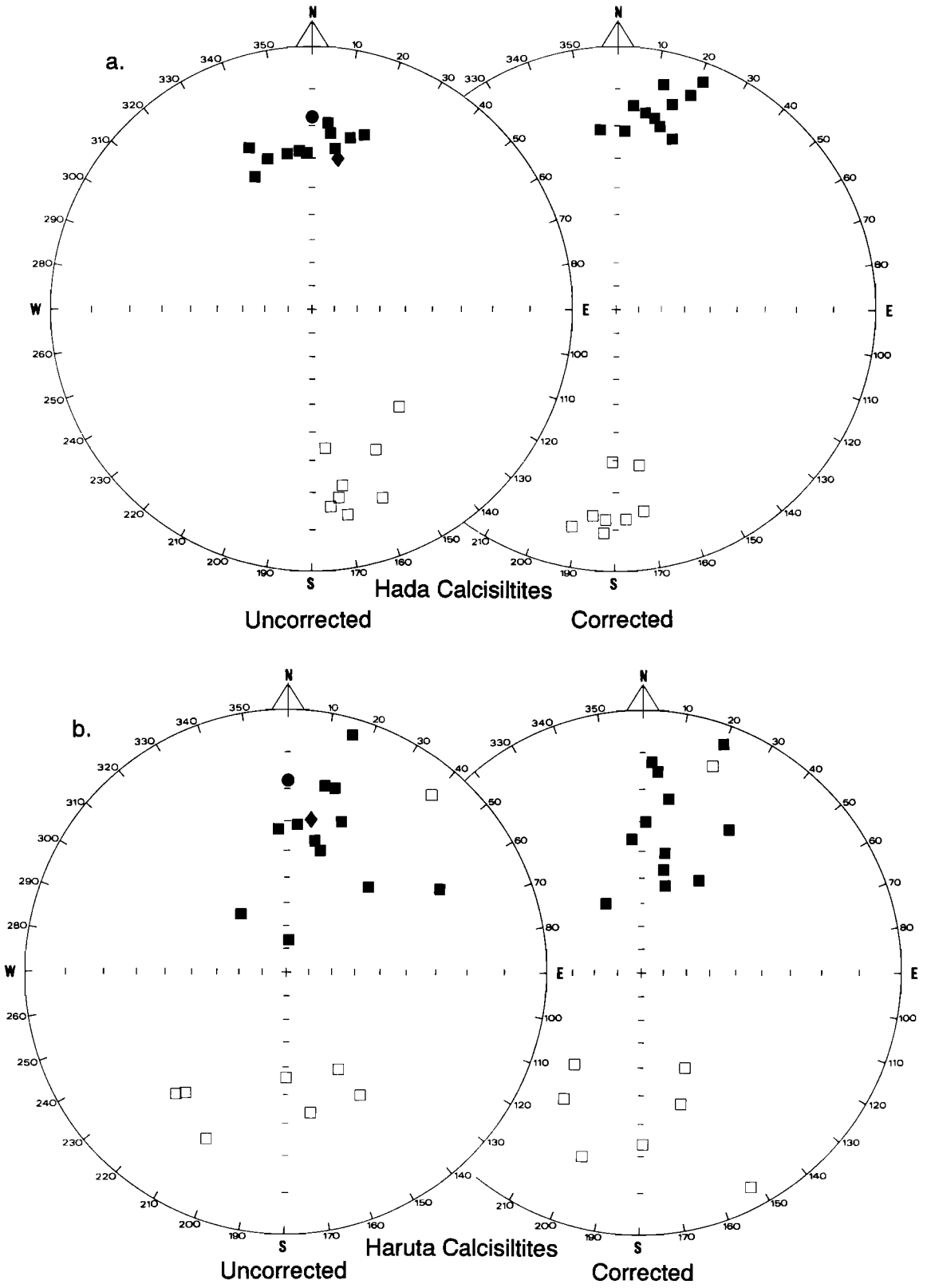
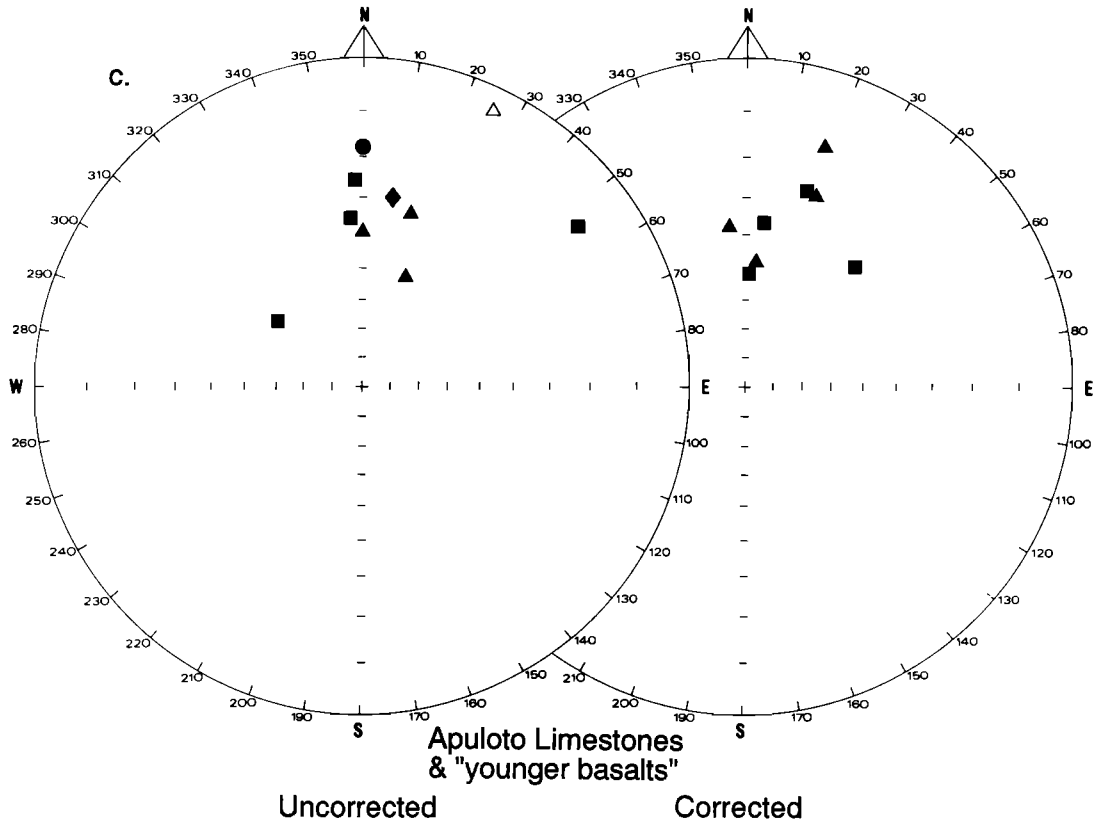


Fig. 11. Stereographic projections of site mean directions (squares) before and after structural correction. Plots of uncorrected directions include the present field (diamond) and axial geocentric dipole field (circle). Solid symbols - upper hemisphere; open symbols - lower hemisphere.

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during) the folding of this formation. No likely source of overprinting, other than low-temperature VRM or weathering chemical remanent magnetization (CRM), has affected the Hada Calcisiltites. Weathering CRM, in this rapidly eroded environment, is of very recent origin. The bulk of the deformation in Malaita seems to have been completed by early Pleistocene times, when the Hauhui Conglomerates, which unconformably overlie the 'Are'are Limestones, were deposited [Hughes and Turner, 1976]; it is improbable that a low-temperature VRM would have survived from these times. Instead, the conflicting results of the two tests merely suggests that, while a post-folding VRM overprint is present in significant amounts, the primary magnetization still dominates to the extent that the structural correction makes significant improvements to the grouping of the mean directions.

Given the difficulty in isolating a normal primary direction from the normal VRM overprint in the Hada Calcisiltites, it is likely that the unresolved overprint resides mainly in the normally polarized sites. As the dip of the majority of sites in this formation is similar (around 30° to 50° to the south-west), a systematic bias to the rock unit mean will persist after the application of the structural correction. Reversely polarized specimens are clearly identifiable, and it is unlikely that the VRM overprint will have resulted in any incorrect determinations of site polarity.

Magnetostratigraphy

The preceding analysis indicates that, at least for the Hada and Haruta Calcisiltites, the magnetic polarity of most sites is confidently known. Combining this information with the foraminiferal ages of sites, and their measured stratigraphic relationships, it is possible to generate a magnetostratigraphic column for the Hada Calcisiltites, including the uppermost part of the Haruta Calcisiltites (Figure 12). Interruptions to the stratigraphy within the Haruta Calcisiltites, and the small number of sites sampled within the Apuloto Limestones, prevented the extension of the magnetostratigraphy much below the Haruta-Hada transition. Age limits of polarity intervals were based on the revised Neogene polarity time scale

TABLE 3. Comparison of Structurally Corrected and Uncorrected Unit Mean Directions

	Dec (°E)	Inc (°)	N	α95 (°)
Hada uncorrected	352.0	-24.8	31.4	6.1
Hada corrected	006.6	-15.5	56.2	4.5
Haruta uncorrected	014.4	-31.9	7.3	13.0
Haruta corrected	007.6	-30.5	11.4	10.1
Apuloto uncorrected	005.5	-37.1	4.6	48.2
Apuloto corrected	017.6	-39.1	21.3	20.4
Younger basalts				
uncorrected	016.1	-30.4	10.1	30.4
corrected	010.1	-32.2	20.1	21.0
Apuloto plus basalts				
uncorrected	011.3	-33.7	7.1	22.3
corrected	013.7	-35.7	22.2	12.0

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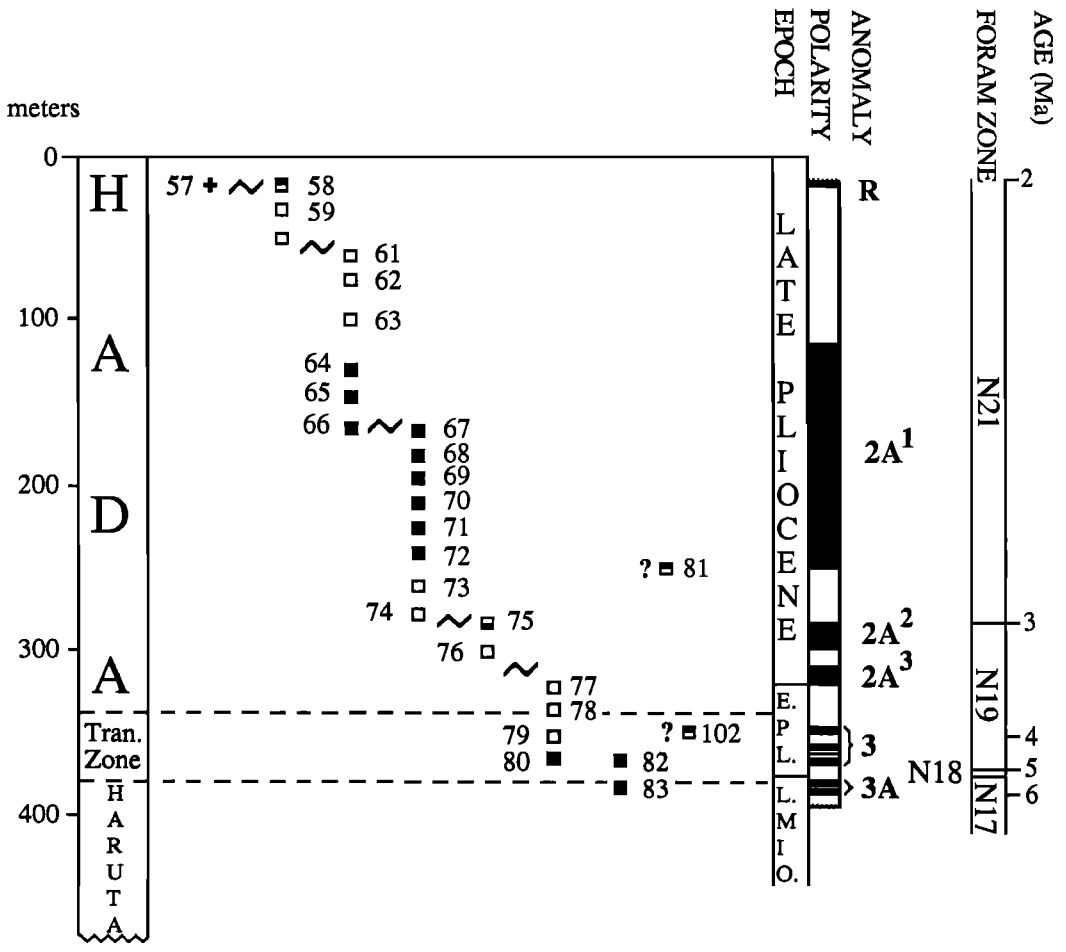


Fig. 12. Magnetostratigraphy of Hada Calcisiltites and uppermost Haruta Calcisiltites. Polarity intervals and foraminiferal zones follow Berggren et al. [1985a]. R marks the Reunion event. Continuously measured sequences are arranged in vertical columns.

of Berggren et al. [1985a]. Gaps between individual measured sections introduce some uncertainties in stratigraphic position, but these uncertainties do not appear to exceed about 20 m within the sequence from SI 57 to SI 80. Despite the apparent lithological uniformity of the Hada Calcisiltites, sedimentation rates varied substantially through the sequence, increasing to a peak between about 3 Ma and 2.5 Ma and declining somewhat before the deposition of the youngest preserved part of the sequence at about 2 Ma. The densely sampled section from SI 64 to SI 72 is restricted to normal interval $2A^1$, implying an average sedimentation rate over this interval of about 290 m/Ma. The base of the Reunion event in anomaly 2 may be represented by the suspected normal polarity at the top of SI 58, implying a sedimentation rate of about 230 m/Ma over the interval from SI 53 to SI 59. Sedimentation rates at the beginning of the Haruta-Hada transition were not much greater than the average for the Haruta sequence, approximately 15 m/Ma based on a thickness of 550 m for the Haruta Calcisiltites [Hughes and Turner, 1976].

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Normally polarized site SI 82 lies within the Haruta-Hada transition zone and is biostratigraphically restricted to either anomaly 3 or the lower part of anomaly 2. Also normally polarized, SI 83 lies 15 m below SI 82 in a well-measured section, but appears to be clearly part of the Haruta Calcisiltites. Given a sedimentation rate of about 15 m/Ma at this time, SI 83 is restricted to anomaly 3A. Initiation of the facies change in this locality can therefore be dated to no earlier than about 5.8 Ma, in the upper part of foraminiferal zone N17. Resig et al. [1986] noted that the first appearance in south Malaita of volcanogenic detritus distributed throughout the sequence (rather than concentrated into mudstone bands) occurs in rocks bearing N17 faunas, confirming this age for the beginning of the Haruta-Hada transition.

Other stratigraphic relationships. Site SI 110, located at the base of the Haruta Calcisiltites, has a normal polarity, as does SI 111, within the transition zone from the Apuloto Limestones. Although no date is available from either site, they must be younger than SI 91, which is at least 30 m below SI 111 in the section and has been dated to zone P12 in the middle Eocene. Deposition rates throughout the Apuloto Limestones probably averaged between about 10 and 20 m/Ma, given the net thickness of about 550 m for the formation [Hughes and Turner, 1976], so that SI 111 is probably at least 1.5 Ma younger than SI 91, and probably around 3 to 4 Ma younger. Zone P12 includes anomalies 19 and 20, so the transition from the Apuloto Limestones to the Haruta Calcisiltites probably occurred over either anomaly 17 or 18, corresponding to the latter two-thirds of the Bartonian (late middle Eocene).

Although conventional stratigraphic columns for southern Malaita suggest a series of eruptions of the "younger basalts" throughout late Cretaceous and Eocene times (see Figure 4), Hughes and Turner [1976] do not report any explicit cases where Apuloto Limestones conformably overly flows of the "younger basalts". Rather, where a conformable relationship has been reported, Haruta Calcisiltites overlie flows of the "younger basalts". No field evidence was observed in the paleomagnetic study for extrusion of the lavas earlier than times just before the transition to the Haruta Calcisiltites. The only clear stratigraphic relationship seen in this study is indicated by the intrusion of the dyke sampled at SI 92, which intrudes Apuloto Limestones no more than 100 m stratigraphically below SI 91; a maximum age in the early middle Eocene for the intrusion follows from the average deposition rate of the Apuloto Limestones. Extrusion of the "younger basalts" at, or just before, the Apuloto-Haruta transition would also be consistent with the observation that all of the sampled "younger basalts" are normally polarized, as normal polarities dominated in the later Bartonian. SI 104 was drilled in Apuloto Limestones only a short interval above the "younger basalts". If, as suggested above, SI 106 was drilled in the same sequence of "younger basalts" as SI 107 - SI 109, then SI 105, in Apuloto Limestones in contact with SI 106, is also of a similar age. Therefore, the sampled sequence in the Apuloto Limestones and "younger basalts" is probably restricted to a relatively short interval from about the base of anomaly 20 to the top of anomaly 17 (46.2 Ma to 39.6 Ma, according to the timescale of Berggren et al. [1985b]) with a bias to the younger age range. A mean age of about 42 Ma follows.

Paleontology and stratigraphic relationships indicate that three Haruta sites were sampled in foraminiferal zone N3, six sites were younger than N3, and twelve were older (the position of the remaining site, SI 101, was unclear). Assuming an even distribution of sites within these age ranges, a mean age of 27.5 Ma for the sampled Haruta Calcisiltites follows.

Bias in the polarity ? Normal and reversed sites occur in equal numbers in the Hada section. However, a slight apparent bias exists in the Haruta Calcisiltites, where normal polarity characterizes 60% of the stable sites. Wherever there are an excess of normally polarized sites, with magnetization directions similar to the present field, the suspicion arises that at least some of the supposed primary magnetizations are in fact VRMs. However, in a sample of this size (N = 20) the probability of an imbalance in the distribution of sampled polarities of this size or greater occurring is 50%, and so the bias is not significant.

Viewed in the simple terms of the analyses carried out on the Hada and Haruta Calcisiltites, the uniform normal polarity of all eight stable sites from the Apuloto Limestones and "younger basalts" would seem to indicate overprinting. As the fold test implies that the characteristic magnetization of the Apuloto Limestones and "younger basalts" was acquired prior to or during deformation of the sequence, possible sources of overprinting are restricted. Overprinting can occur when erosion "freezes in" a magnetization acquired at elevated burial temperatures. The characteristic magnetization of the Apuloto Limestones and "younger basalts", which is carried by magnetite, is stable at laboratory demagnetization temperatures of greater than 500°C. Demagnetization temperatures were kept stable for at least 100 seconds. The upper limit of the duration of an overprinting field is set to 5 Ma, as intervals of constant (or predominantly uniform) geomagnetic polarity did not exceed this length during the Tertiary. Applying the thermal activation nomogram for magnetite of Walton [1980] results in an estimate for the burial paleotemperature of at least 300°C. Even if Walton's thermal activation relationship overestimates paleotemperatures, it is very unlikely that adequate paleotemperatures could have applied to produce such an overprint, given that less than 1.5 km of the overlying strata has been removed from the Apuloto Limestones. Thermal overprinting of the Apuloto Limestone by intrusion of the "younger basalts" is a possibility, although it would be surprising in this case that none of the demagnetization paths shows evidence of the original magnetization, or that the overprint should be so widespread. If intrusion of the "younger basalts" were the source of the Apuloto magnetization, then a late middle Eocene age would still be appropriate for this event. Alnoitic intrusion in the early Oligocene is restricted to northern Malaita, and would not have produced any significant thermal effects in the sampled region. No other source of pre-deformation overprint is obvious.

Over the sampled interval of the Apuloto Limestones - "younger basalts" sequence, which is bounded by anomalies 20 to 17, normal polarity intervals predominate, making up about 70% of the total. In addition, the basalt layer sampled by SI 107 - SI 109 and SI 106 is likely to represent a series of closely-spaced extrusive events; in this case, SI 107 - SI 109 and SI 105 effectively represent only a single sample of the polarity. Given these conditions, the effective number of independent samples of the polarity, including SI 111 within the overlying transition zone, is reduced to five. The probability of all five samples having normal polarity in a sequence which is 70% normal is $(0.7)^5$, or about 17%. This estimate is probably conservative; if, as suggested above, the "younger basalts" were erupted just before the Apuloto-Haruta transition in the later Bartonian then all of the sampled Apuloto and "younger basalts" sites with the exception of SI 91 were sampled in the 80% normal interval over anomalies

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17 and 18. Hence, a primary origin for the magnetization in the Apuloto limestones and "younger basalts" is not rejected, and the positive result of the fold test need not be doubted.

Paleomagnetic Poles

Paleomagnetic poles were calculated for each of the four units (Hada, Haruta, Apuloto and "younger basalts"), and for the combined middle Eocene (Apuloto + "younger basalts") by averaging virtual geomagnetic poles (VGPs) determined from the mean magnetization direction for each site (Table 4, Figure 13). The Apuloto and "younger basalts" data sets are small, and their combination, given their similar age, is legitimate. The Hada pole was based on reversed data only.

To understand the tectonic significance of the paleomagnetic poles, reference apparent polar wander paths (APWPs) for the Australian and Pacific plates are needed. Neither APWP is unambiguously defined, however. Several versions of the Australian APWP exist. While the trend of all recent Australian Tertiary APWPs has been towards a relatively simple path, largely defined by laterite data and directed roughly along the 120°E meridian, there is considerable disagreement over the rate of pole motion. Paths using data from igneous bodies from eastern Australia to calibrate for age [Embleton, 1981; Embleton and McElhinny, 1982; Musgrave and Falvey, 1983; Musgrave, 1989] indicate a roughly linear rate of apparent polar motion through the middle to late Tertiary, in agreement with the absolute motion of Australia implied by hot-spot data [Wellman, 1983]. Idnurm [1985] produced a path which showed large variations in the rate of movement since the early Tertiary. The Musgrave APWP is used here; however, as the trajectory of the Idnurm APWP over the middle and late Tertiary is similar, tectonic arguments based on one path will not differ greatly from those based on the other. There will, however, be some disagreement concerning dating.

Seamount poles form the largest database from which the Pacific APWP can be defined. Despite controversy over the reliability of these poles, Sager [1987] has shown that they cluster around the approximate Pacific APWP shown in Figure 13. Calibration ages on this path are augmented by poles from Gordon [1983].

In the accepted tectonic history of the region, Malaita (unlike the Central Province) should always have been part of the Pacific plate. Comparison of the Hada, Haruta and combined Eocene poles with the two reference APWPs implies otherwise, however. Both the pre-Pliocene poles lie close to the Australian APWP, and are nearly distinct at their

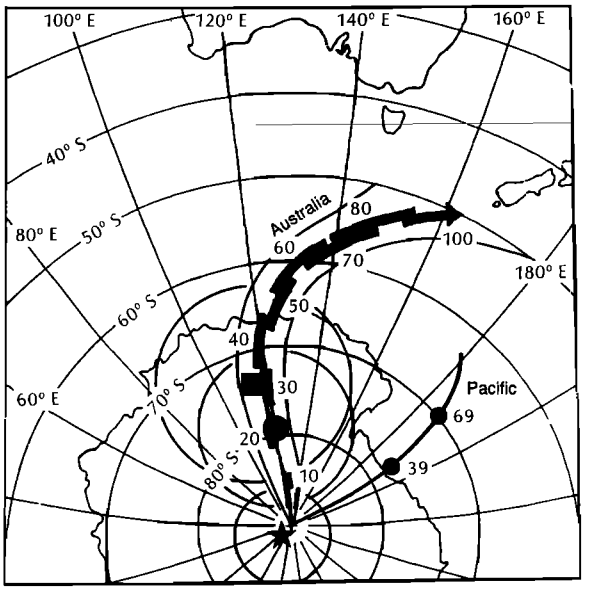


Fig. 13. Paleomagnetic poles from Malaita: star, Hada Calcisiltites; circle, Haruta Calcisiltites; square, combined Eocene (Apuloto Limestones plus "younger basalts"). Circles of 95% confidence are shown around each pole. Australian APWP is shown flanked on alternating sides by 95% confidence limits of position along the path of the indicated ages in Ma [cf. Musgrave, 1989]. Lines bounding Australian APWP on both sides are 95% confidence limits of the trajectory of the APWP. Pacific APWP [Sager, 1987] is shown as a single line to the east of the Australian APWP, with dots showing position of 39 and 69 Ma poles [Gordon, 1983].

95% confidence level from the Pacific APWP. The Pliocene Hada pole may be associated with either reference path.

Anticlockwise rotation of the Eocene and Haruta poles through about 15° will bring them into agreement with the Pacific APWP within their confidence limits, although the positions of the rotated poles do not closely match the Pacific path (Figure 14). Similar rotation of the Hada pole, however, displaces this pole well away from the Pacific APWP. Six of the seven reversely polarized stable sites from which the Hada pole is drawn were deposited before 2.9 Ma, and so before the

TABLE 4. Paleomagnetic Poles from Malaita

Unit	Mean Age (Ma)	Latitude (°N)	Longitude (°E)	N	α95 (°)
Hada Calcisiltites - reversed sites only	3	-88.5	007.1	8	5.5
Haruta Calcisiltites	23	-79.2	120.3	20	8.9
Apuloto Limestones	42	-68.9	112.5	4	21.6
Younger basalts	42	-77.4	116.3	4	17.5
Combined Apuloto Ls and younger basalts	42	-73.2	114.0	8	11.4

All poles are given in their southern hemisphere projection.

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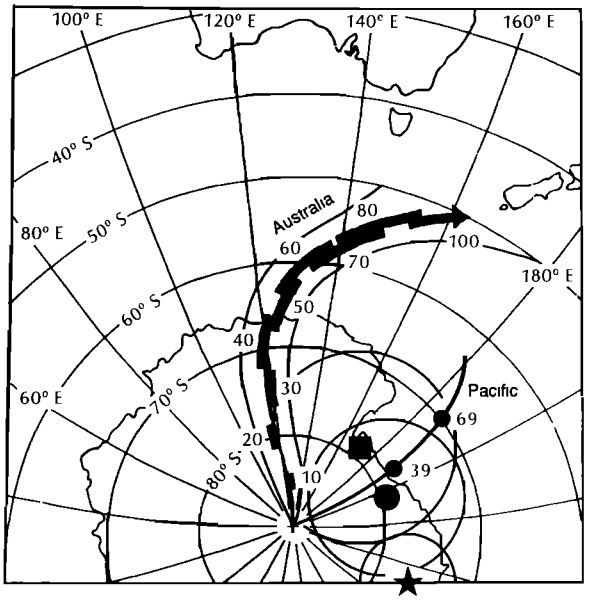


Fig. 14. Malaitan paleomagnetic poles after anticlockwise rotation of 15° about a local vertical axis. Symbols as for Figure 13.

beginning of uplift of the Pacific Province [Resig et al., 1986]. Tectonic rotation at or after the time of the deformation of the Pacific Province is therefore precluded. It seems unlikely that significant tectonic rotation of a block within the thick crust of the Ontong Java Plateau could have occurred without accompanying deformation, and so tectonic rotation cannot explain the displacement of the Eocene and Haruta poles from the Pacific APWP.

This result is incompatible with the standard model. It indicates that for at least most of the time since the late Eocene Malaita has been on the Australian plate, with at most a few million years of motion in concert with the Pacific plate. At most, the standard model would allow a few million years in which Malaita was on the Australian plate, between the time it was overthrust onto the Solomons Arc and the arc polarity reversal, both events occurring in the late Miocene; this is not sufficient to explain the strongly "Australian" trend of the two older poles. Although Kroenke [1984] places the arc polarity reversal earlier in the Miocene, his model fares no better, as it still leaves Malaita on the Pacific plate from at least the middle Eocene.

If Malaita (and by inference, the rest of the Pacific Province) has been on the Australian plate for most of its history, then it may be expected to have followed the tectonic history usually outlined for the Central Province, including the arc polarity reversal. Pre-Pliocene poles from Malaita can be directly compared with the Australian APWP if they are restored to the position they would have had if the arc polarity reversal had not taken place, and the Solomons Arc had remained on the Australian plate. This is achieved by rotating the poles around the Pacific-Australia finite reconstruction pole for the time of the polarity reversal. Pacific-Australia reconstruction poles were calculated from proportions of the Pacific-Antarctic reconstruction poles for 3.2 Ma [Minster and Jordan, 1978; Cox and Engebretson, 1985; Pollitz, 1986] and 9.8 Ma [Stock and Molnar, 1982] and the Australia-Antarctic pole for 9.8 Ma [Stock and Molnar, 1982]. The closest restoration of the two poles to their corresponding positions on the Australian APWP was achieved by a reconstruction pole for 6.5 Ma (Figure 15). The agreement with the Australian APWP is very close. Both poles lie well within the confidence limits of the trajectory of the Australian path. Both poles also lie close to their corresponding age on the Australian APWP, and there is extensive overlap of the confidence limits of the poles and their corresponding points on the APWP. The Pacific path, on the other hand, lies entirely outside the 95% confidence limits of the two poles. Strong support is indicated for the hypothesis that Malaita lay on the Australian plate prior to being transferred to the Pacific plate during the arc reversal, and had not formerly been part of the Pacific plate since the late Eocene at least.

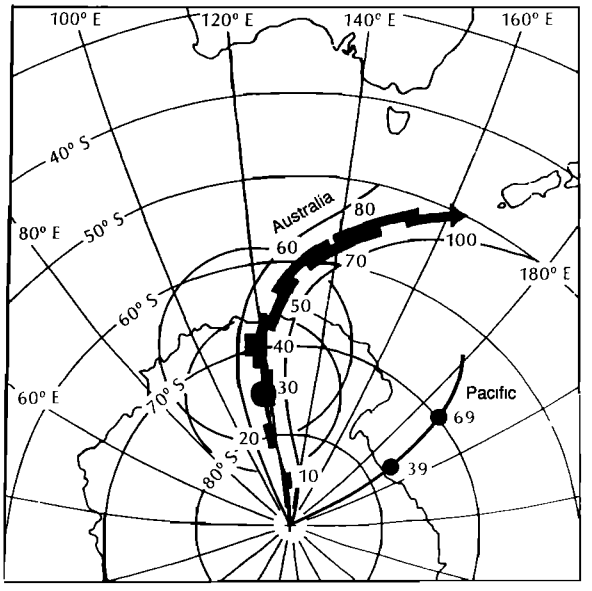


Fig. 15. Haruta Calcisiltites (circle) and combined Eocene (square) poles after restoration for 6.5 Ma of Pacific-Australia motion.

Paleolatitudes and the Ontong Java Plateau

Hammond et al. [1975] produced a paleomagnetic paleolatitude curve for DSDP site 289 on the OJP which showed a fairly constant rate of decrease in paleolatitude from 70 to 30 Ma, with no further change indicated from 30 Ma to the present. Unrecognized spreading between the OJP and the Pacific Plate may have allowed the OJP to remain at a fixed latitude; convergence between the OJP and the Solomons Arc would still be required. Paleolatitudes calculated from the Malaitan paleomagnetic data lie distinctly closer to a paleolatitude curve which assumes that Malaita transferred from the Australian to the Pacific plate at about 6.5 Ma than to the OJP paleolatitude curve (adjusted to match the current position of Malaita relative to DSDP site 289) (Figure 16), although both curves lie within two standard deviations of the data.

DISCUSSION

Lithological, metamorphic, and structural differences between the Pacific and Central Provinces do not require that

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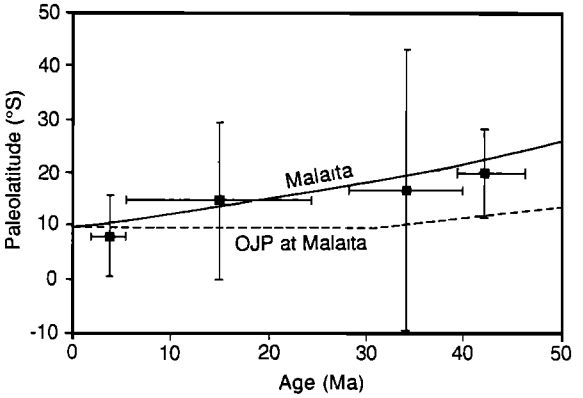


Fig. 16. Magnetic paleolatitude data from Malaita compared to paleolatitude curves assuming Malaita transferred from the Australian to the Pacific plate at 6.5 Ma ("Malaita") and that Malaita originated as part of the OJP ("OJP at Malaita"). Two means are derived from the Haruta Calcisiltites, one for sites younger than foraminiferal zone N3, the other for sites older than N3. Error bars indicate likely age ranges for data sets and two standard deviations of the mean paleolatitudes.

these two units arose in remote locations. The argument that Malaita is a fragment of the OJP largely rests on the supposed stratigraphic similarity of their sequences [Kroenke, 1972; Packham and Andrews, 1975; Hughes and Turner, 1976]. However, any similarities may simply reflect the development of two Tertiary bathyal marine sequences in tropical or near-tropical latitudes. Both sequences dominantly comprise planktonic nannofossil and foraminiferal debris, they have similar basement ages, and both have cherts near their base. Kroenke [1972] suggested that seismic reflectors observed on the plateau may be lava flows or sills correlating with the "younger basalts" of Malaita, but there is no direct evidence to support this. Notable differences between the two sequences exist: the mudstone turbidites of the Haruta Calcisiltites are lacking on the OJP, and a major sedimentary hiatus in the Eocene and lower Oligocene of the OJP is absent from the Malaitan succession. While these differences might only reflect local facies variation over the more than 300 km between Malaita and the nearest DSDP site, the similarities are not sufficient to support an association of Malaita and the OJP.

Continuity of the pre-Pliocene sequence from the OJP to Malaita, and hence a genetic linkage between the Pacific Province and the OJP, has also been inferred from seismic lines crossing the North Solomons Trench (Figure 17) [Kroenke, 1972; Bruns et al., 1986b]. Seismic reflectors can be continuously traced across the OJP and, on at least one section (S-S', Figure 17), onto the outer part of the Malaita Anticlinorium [Kroenke, 1972]; these cannot be continued further south on the anticlinorium or onto Malaita itself, however, although Kroenke related these reflectors to the major divisions within the Malaitan sequence. Other profiles shown by Kroenke [1972] to both the west and east of S-S' do not indicate an extension of the OJP sedimentary cover beyond the North Solomons Trench. Line 37 of Bruns et al. [1986b] shows 1 km of sediments overlying basement continuing across the North Solomon Trench. Reflectors in this sequence cannot be traced beneath the bathymetric high which bounds the trench to the south, and so the trench here does not mark the surface expression of a subduction zone. However, the reflectors cannot be continually traced across the Malaita Anticlinorium, so a boundary between OJP crust and the Pacific Province may locally lie within the anticlinorium. It is significant that these two seismic profiles cross the North Solomon Trench where it becomes much narrower and shallower. Significant deformation may extend here into the OJP crust, so that the suture between the OJP and the Solomons Arc lies within the anticlinorium. Both NW and SE of this region the North Solomon Trench is broader and deeper, and the anticlinorium less developed; the suture in these regions probably lies at, or close to, the south-western wall of the trench. A series of troughs traces the probable line of the suture within the broader part of the anticlinorium.

Seismicity which indicates that the south-west dipping Benioff zone intersects the surface at the North Solomons Trench [Cooper et al., 1986; Cooper and Taylor, 1987] is consistent with the model in which this trench marks the relict subduction zone separating the Pacific Province and the OJP, and is less compatible with the alternate view that the former subduction zone lay between Malaita and the Central Province and is now marked by the KKK fault system. It is difficult to see how the Pacific Province, if it had previously been a part of the OJP, could have been thrust over the North Solomon Trench yet still have preserved the bathymetric expression of the trench.

Foraminiferal paleobathymetric data from Malaita and Ulawa [Resig et al., 1986] indicate that significant uplift of Malaita did not occur at the time of the Haruta-Hada facies change. Instead, uplift was delayed until about 3 Ma and uplift rates peaked at around 1.5 Ma before present. Increased influx of volcanogenic detritus to Malaita and uplift and deformation of the Malaita Anticlinorium may represent discrete stages in a protracted process of arc polarity reversal. It is not necessary that subduction of the Pacific plate must have ceased as soon as subduction of the Australian plate commenced; a period of simultaneous subduction of both plates below the Solomon Islands Arc may have occurred. If this were the case, then an independent Solomons Arc microplate would have had a motion relative to the Australian plate which was some fraction of the total Pacific-Australia convergence during the period of simultaneous dual subduction. The 6.5 Ma Pacific-Australia reconstruction of the Malaitan paleomagnetic poles made in Figure 15 would then represent an equivalent Solomons-Australia relative movement during dual subduction at a slower rate (say, half the total Pacific-Australia convergence), beginning rather earlier (at, say, about 10 Ma) and ending later (at, say, about 3 Ma).

Collision between the two downgoing slabs may finally have ended subduction of the Pacific plate, or alternatively, Pacific subduction may have slowed and then halted of its own accord as thicker and so more buoyant OJP crust was forced into the subduction zone. Subduction of the Australian plate may have brought about a renewal of volcanism in the Central Province, and accompanying tectonism may have reduced the effectiveness of the Korighole-Florida High [Ramsay, 1982] as a barrier to sediment transport. Increased terrigenous sedimentation in Malaita presumably represents an intermediate stage in the polarity reversal, when sufficient of the Australian plate had been consumed to engender volcanism. After Pacific subduction halted, completing the arc polarity reversal, remaining convergence between the Pacific plate and the arc may have resulted in the partial detachment of the unsubducted parts of the OJP from the subducted Pacific lithosphere and the collision of the unsubducted OJP with the salient on the Solomons Arc formed by Malaita, producing the uplift and deformation of the Malaita Anticlinorium.

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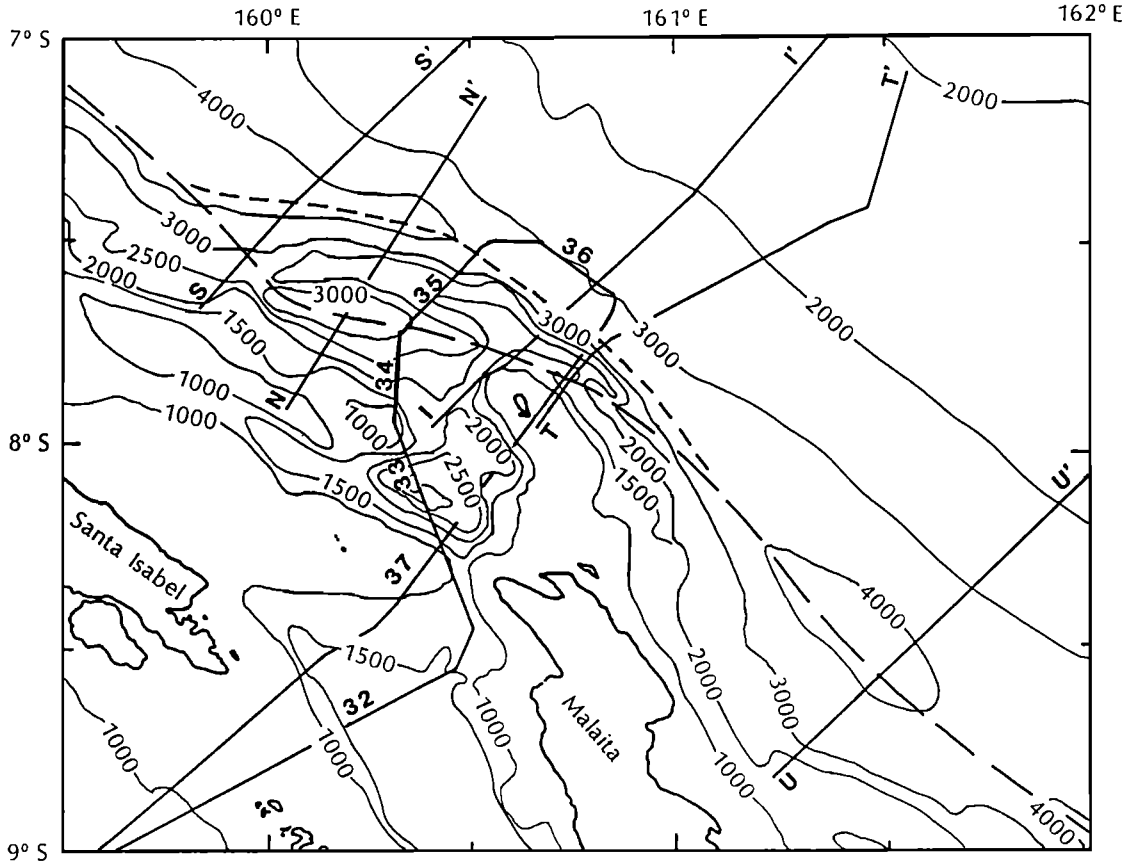


Fig. 17. Positions of seismic lines crossing the North Solomons Trench near Malaita. Lines I-I', N-N', S-S', T-T' and U-U' are reported in Kroenke [1972]; lines 32 to 37 are reported in Bruns et al. [1986a, 1986b]. Bathymetry in metres. Line of long dashes indicates location of former axis of North Solomon Trench, prior to deformation of the Malaita Anticlinorium; line of short dashes follows deepest part of modern trench, usually misidentified as axis of former trench.

The tectonic evolution of the Solomon Islands Arc since the Late Eocene is sketched in Figure 18. The Pacific Province was an integral part of the earlier north-east facing arc, filling the role of the "missing" fore-arc zone. The absence of features suggestive of an accretionary prism suggests tectonic erosion [Hussong et al., 1976] along the old trench, at least during the period immediately preceding arc reversal. Tectonic erosion might have been related to the subduction of the outer, less thick parts of the OJP, and may have been sufficiently active to remove the outer part of the fore-arc in the west of the arc, resulting in the westward termination of the Malaita Anticlinorium.

Metamorphism and uplift in the Central Province accompanied the increase in terrigenous detritus in the change from Apuloto to Haruta sedimentation. Extrusion of the "younger basalts" may have been restricted to a relatively short period just before the Apuloto-Haruta transition. These three features may reflect the onset of subduction of the Pacific plate below the arc, which was the result of the change in Pacific plate absolute motion at anomaly 18 time (42-44 Ma) [Clague and Jarrard, 1973]. Australian-Pacific relative motion before this time was probably sub-parallel to the arc, which would have formed a transform boundary prior to the late Eocene. Terrigenous input to the Haruta Calcisiltites was low, suggesting that the arc was not highly volcanically productive, perhaps due to highly oblique subduction following the change in Pacific plate motion. Oblique subduction may also have given rise to sinistral strike-slip motion within the arc along the KKK fault system.

Subduction of the margins of the OJP began in the late Miocene; resistance to subduction of this buoyant feature locally slowed Australia-Pacific plate convergence, until stress was relieved by the initiation of subduction of the Australian plate below the south-western side of the arc. Uplift and tectonism in Guadalcanal at this time [Hackman, 1980] may reflect this event. By 5.8 Ma subduction of the Australian plate was being accompanied by renewed volcanism, and the deposition of the Hada Calcisiltites followed. Opening of the Woodlark Basin commenced at least as early as anomaly 3 time [Taylor, 1987], and may also have resulted from complex tectonics related to the early phases of Australian plate subduction. Simultaneous subduction of the Pacific, Australian and Solomon Sea plates continued until about 3 Ma, when subduction of the Pacific plate broke down, and deformation and uplift of the Malaita Anticlinorium followed as the unsubducted part of the OJP was forced against the old fore-arc. Uplift and deformation continued until about 1 Ma accompanying collision between the OJP and the Solomons

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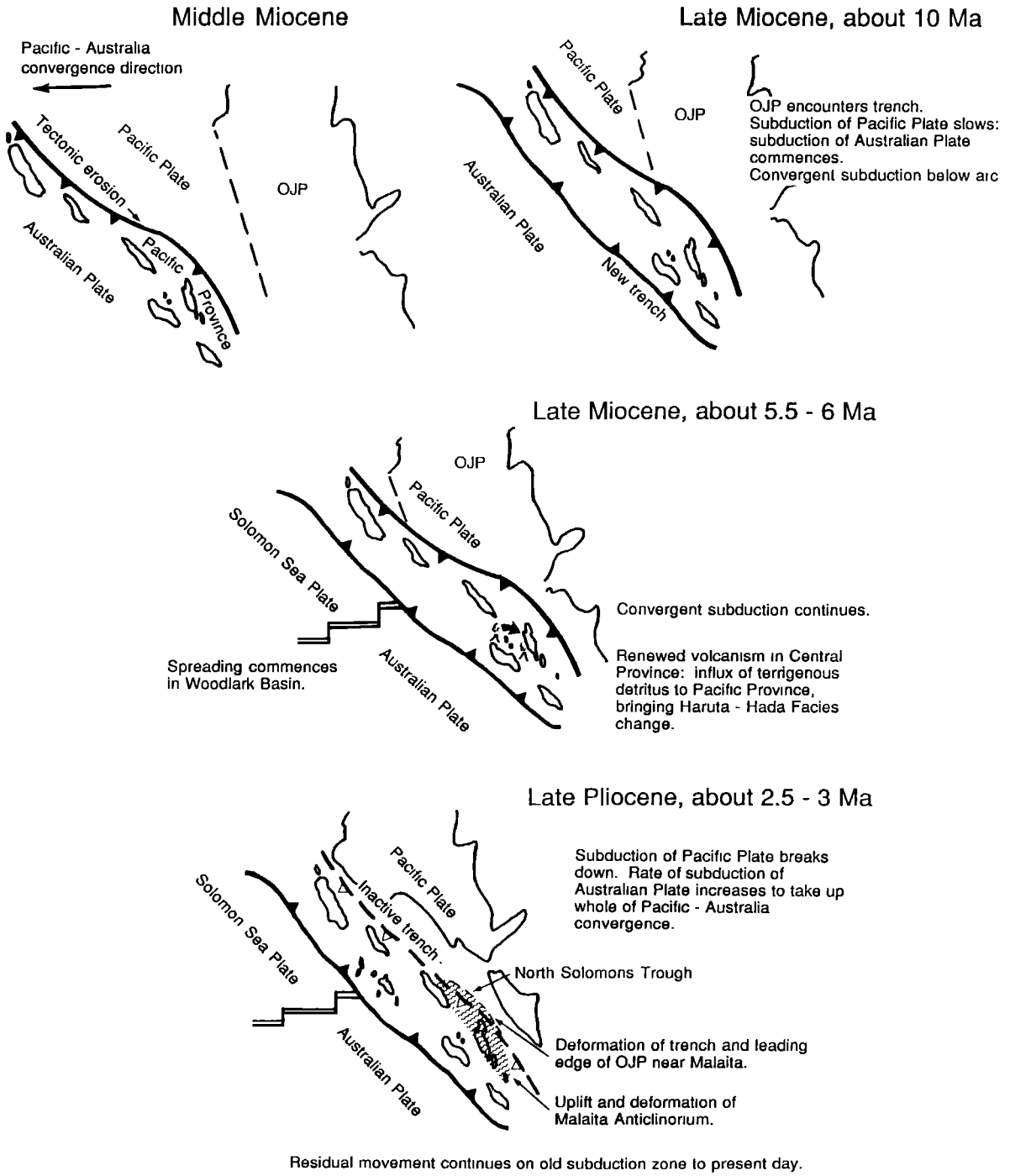


Fig. 18. New model for the tectonic evolution of the Solomon Islands Arc since the middle Miocene.

Arc while subduction of the Australian and Solomon Sea plates adjusted to accommodate the entire local Pacific-Australia convergence.

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